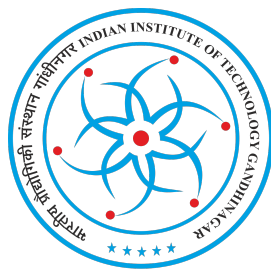


INDIAN INSTITUTE OF TECHNOLOGY GANDHINAGAR (IIT Gandhinagar)



Assignment Report

M.Tech in Integrated Circuit Design and Technology (ICDT)

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Subject Name:	VLSI Design
Subject Code:	EE 617
Assignment No.:	02
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Question 1

[Marks: 60] For each of the following cases calculate the delay, static power and dynamic power for a CMOS inverter for the below combinations. You need to ensure that for each case, the rise and fall delays are equal. Also, find the NML and NMH for each case, and explain your observations. Show the results on a single plot for each case.

(a) Varying supply voltages – 3 values

Table 1: Delay, Static Power, Dynamic Power, and Noise Margins for different V_{DD} values

Parameter	Vdd = 1.8V	Vdd = 1.4V	Vdd = 1.0V
W_n (nm)	0.22	0.22	0.22
W_p (nm)	0.9987	1.2	1.45
Rise Delay (ps)	25.91	35.28	55.60
Fall Delay (ps)	26.81	37.96	60.50
P_{stat} Vin=0 (pW)	9.82	6.44	3.842
P_{stat} Vin=Vdd (pW)	9.51	8.31	6.294
P_{dyn} LH (μ W)	100.549	61.43	35.17
P_{dyn} HL (μ W)	96.151	56.31	29.72
NML (mV)	759.348	788.018	539.877
NMH (mV)	773.47	781.9	555.625

- **Delay:** As V_{DD} decreases, rise and fall delays increase because transistors switch more slowly.
- **Static Power:** Static power reduces at lower V_{DD} , since leakage current is smaller.
- **Dynamic Power:** Dynamic power drops exponentially with lower V_{DD} , as it depends on V_{DD}^2 .
- **Noise Margins:** Both NML and NMH reduce when V_{DD} is lowered, giving weaker noise margin.

(1) Vdd = 1.8 , Ln = 0.18nm , Wn = 0.22nm , Lp = 0.18nm , Wp = 0.9987nm

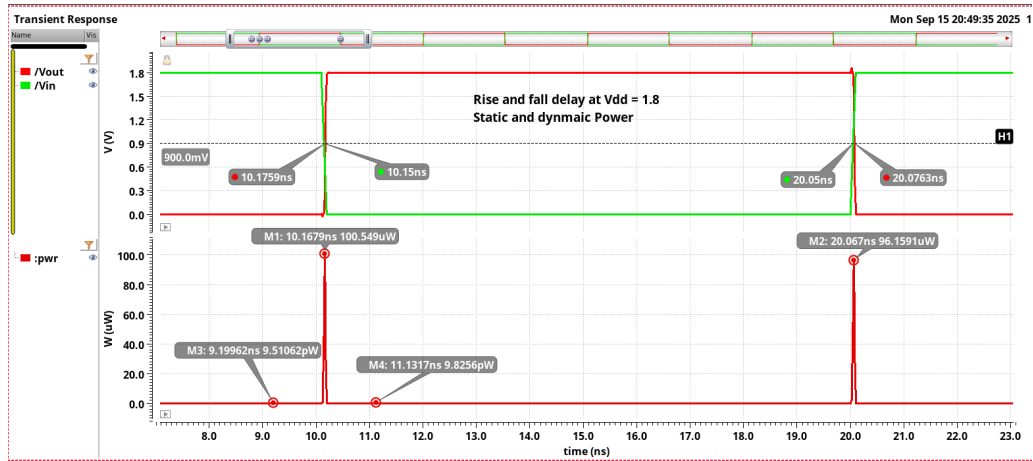


Figure 1: Rise and fall delay and Static and dynamic power at Vdd = 1.8V

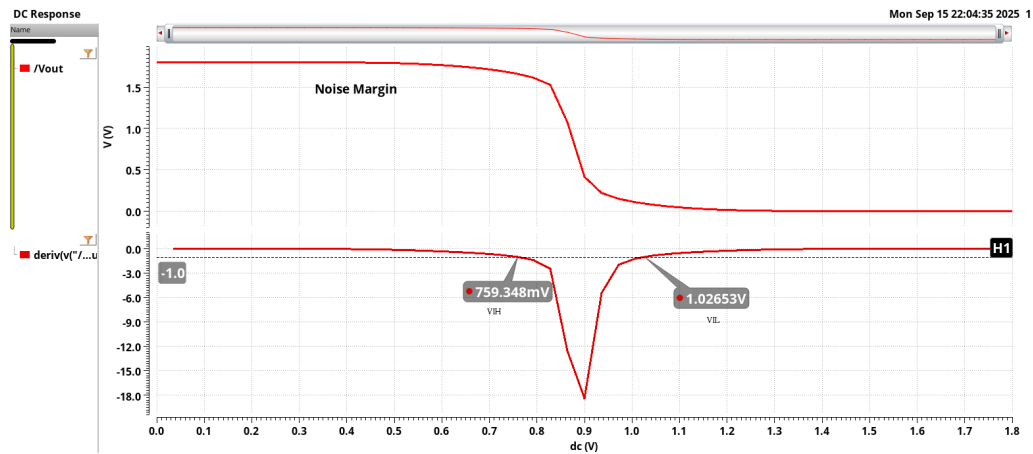


Figure 2: Noise margin at Vdd = 1.8V

(2) Vdd = 1.4 , Ln = 0.18nm , Wn = 0.22nm , Lp = 0.18nm , Wp = 1.2nm

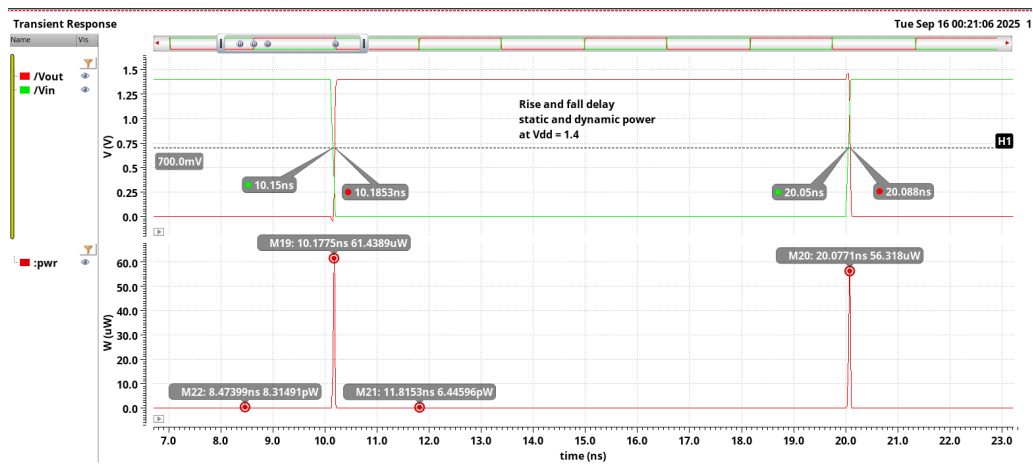
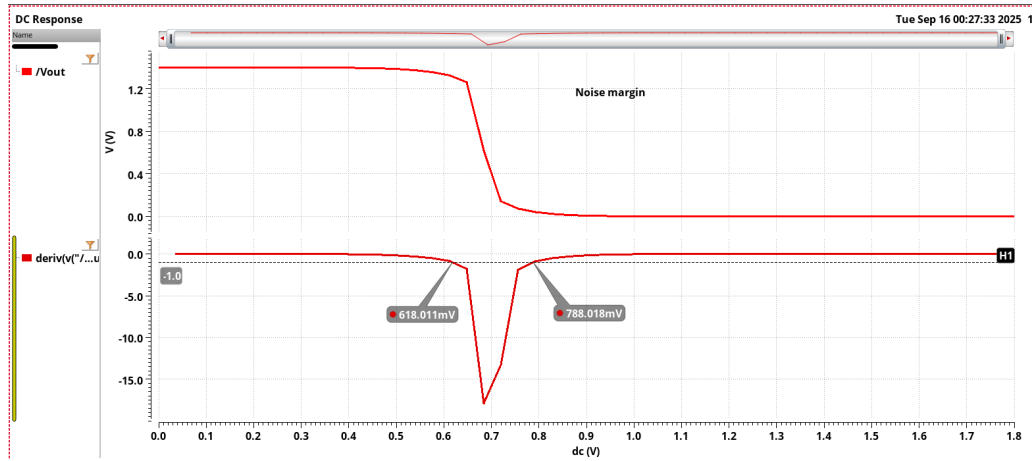
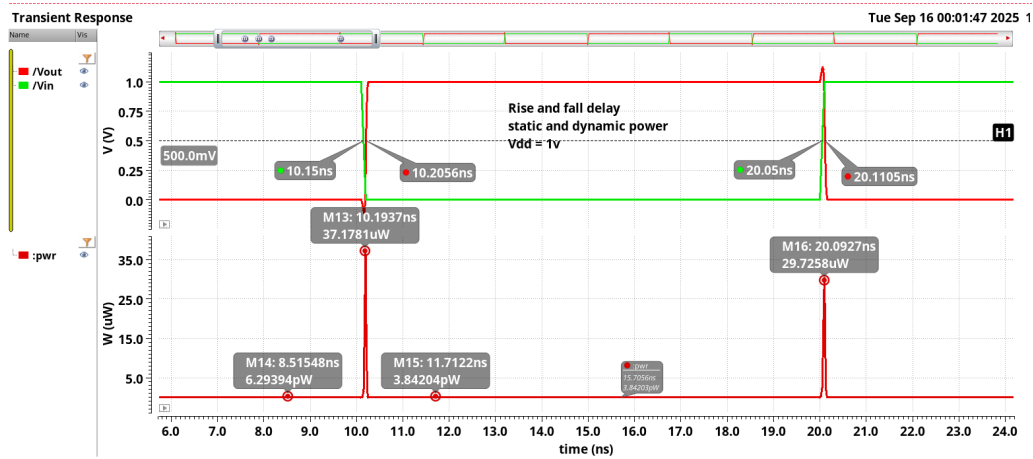
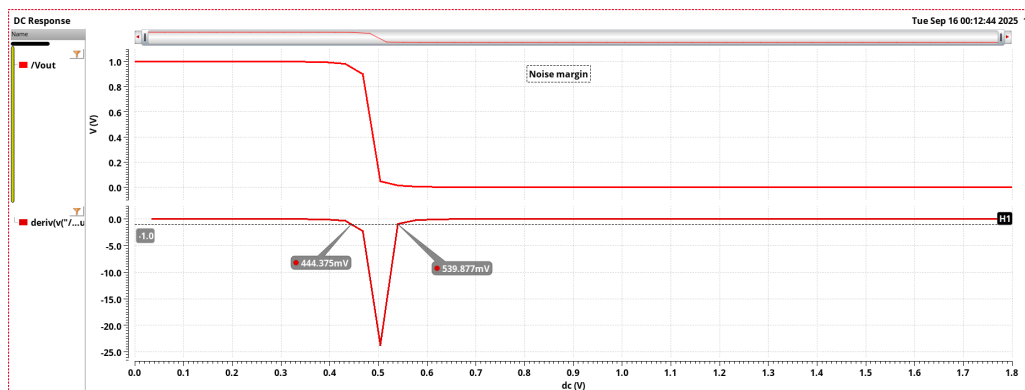


Figure 3: Rise and fall delay and Static and dynamic power at Vdd = 1.4V

Figure 4: Noise margin at $V_{dd} = 1.4V$

(3) $V_{dd} = 1.0$, $L_n = 0.18nm$, $W_n = 0.22nm$, $L_p = 0.18nm$, $W_p = 1.45nm$

Figure 5: Rise and fall delay and Static and dynamic power at $V_{dd} = 1.0V$ Figure 6: Noise margin at $V_{dd} = 1.0V$

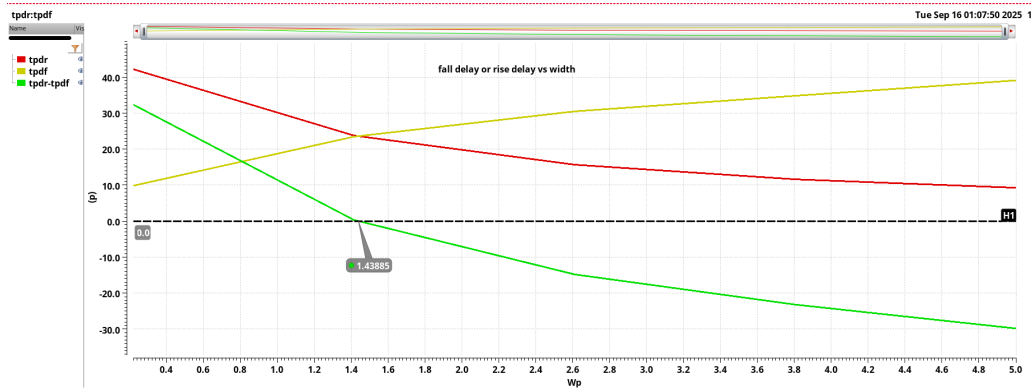
(b) Varying device widths for minimum $L - 3$ values

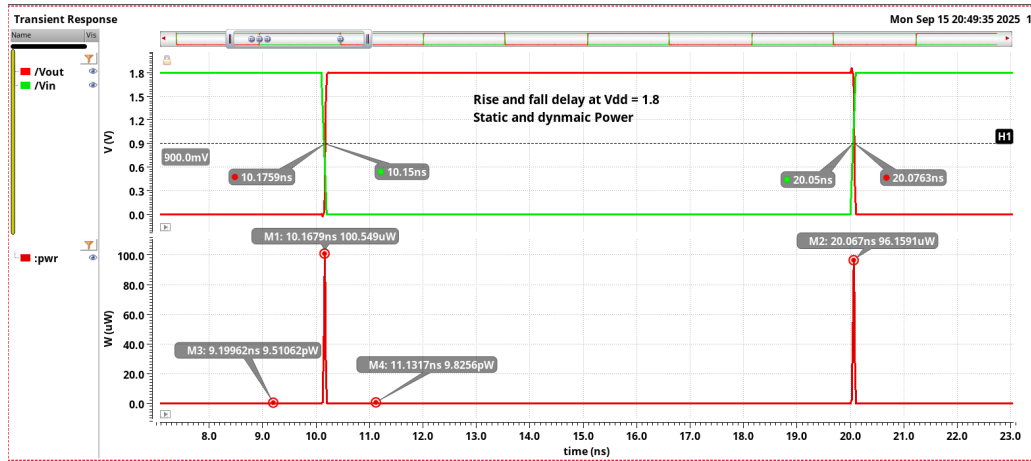
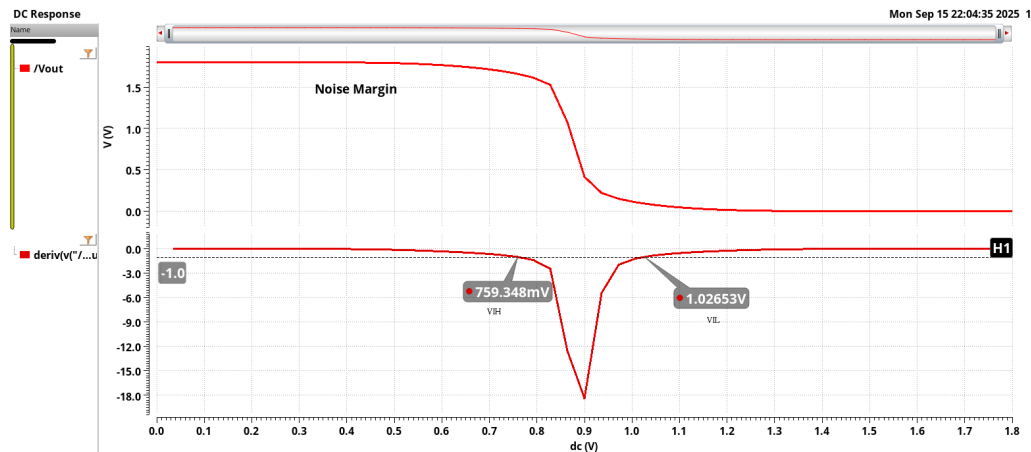
Table 2: Delay, Static Power, Dynamic Power, and Noise Margins for different W_n and W_p values at $V_{DD} = 1.8V$

Parameter	Case 1	Case 2	Case 3
V_{DD} (V)	1.8	1.8	1.8
W_n (nm)	0.22	0.44	0.66
W_p (nm)	0.9987	1.43885	1.96038
Rise Delay (ps)	25.91	23.52	21.04
Fall Delay (ps)	26.81	23.72	21.61
P_{stat} $V_{in}=0$ (pW)	9.82	3.16	3.415
P_{stat} $V_{in}=V_{dd}$ (pW)	9.51	16.30	24.78
P_{dyn} LH (μW)	100.549	149.751	206.532
P_{dyn} HL (μW)	96.151	149.554	206.371
NML (mV)	759.348	1029.900	1036.330
NMH (mV)	773.470	1032.697	1028.825

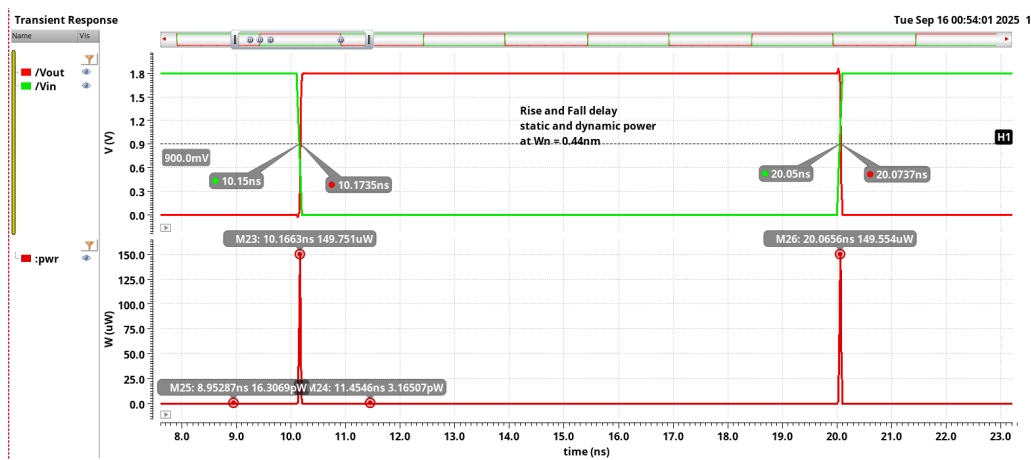
- **Delay:** Rise and fall delays reduce as W_n and W_p increase, because wider transistors provide stronger drive strength.
- **Static Power:** At $V_{in} = 0$, static power decreases slightly, while at $V_{in} = V_{DD}$, it increases with larger widths due to leakage.
- **Dynamic Power:** Both LH and HL dynamic power increase steadily with width.
- **Noise Margins:** NML improves with higher widths, and NMH also increases initially but shows constanst at larger sizes.

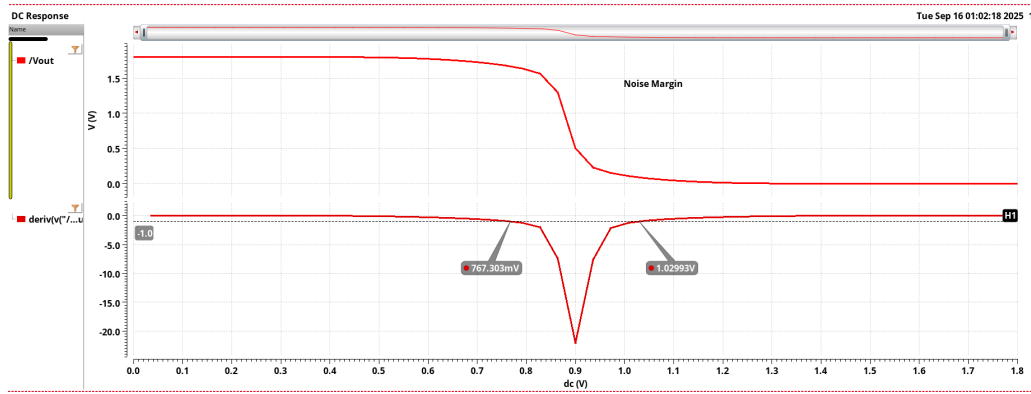
(1) $V_{dd} = 1.8$, $L_n = 0.18nm$, $W_n = 0.22nm$, $L_p = 0.18nm$, $W_p = 0.9987nm$

Figure 7: Calculation of W_p to make delay equal

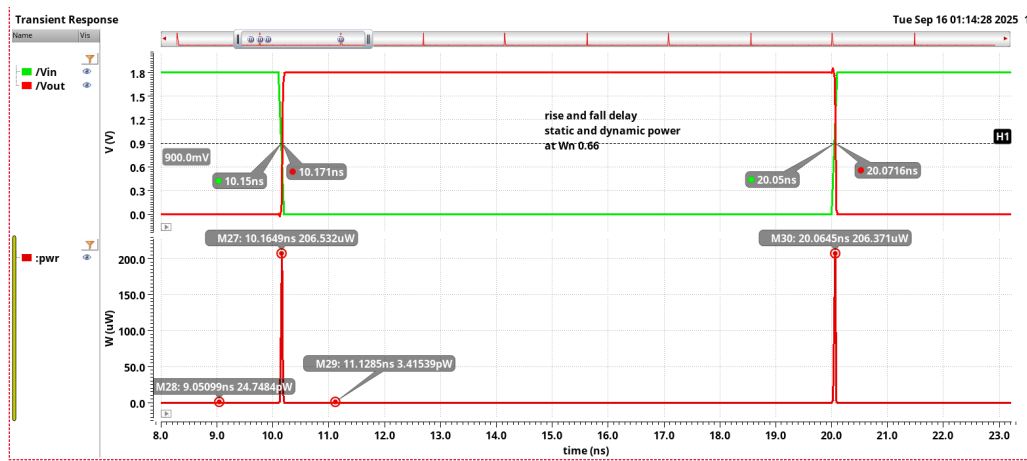
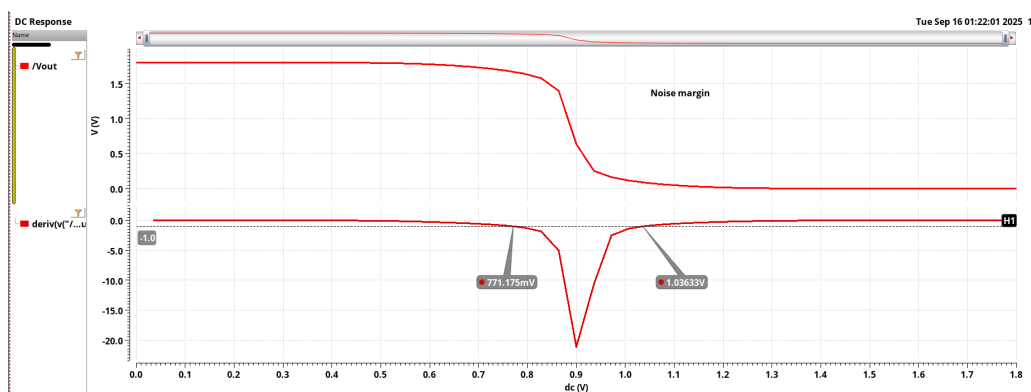
Figure 8: Rise and fall delay and static/dynamic power at $W_n = 0.22\text{nm}$ Figure 9: Noise margin at $W_n = 0.22\text{nm}$

(2) $V_{dd} = 1.8$, $L_n = 0.18\text{nm}$, $W_n = 0.22\text{nm}$, $L_p = 0.18\text{nm}$, $W_p = 1.43885\text{nm}$

Figure 10: Rise and fall delay and static/dynamic power at $W_n = 0.44\text{nm}$

Figure 11: Noise margin at $W_n = 0.44\text{nm}$

(3) $V_{dd} = 1.8$, $L_n = 0.18\text{nm}$, $W_n = 0.22\text{nm}$, $L_p = 0.18\text{nm}$, $W_p = 1.96038\text{nm}$

Figure 12: Rise and fall delay and static/dynamic power at $W_n = 0.66\text{nm}$ Figure 13: Noise margin at $W_n = 0.66\text{nm}$

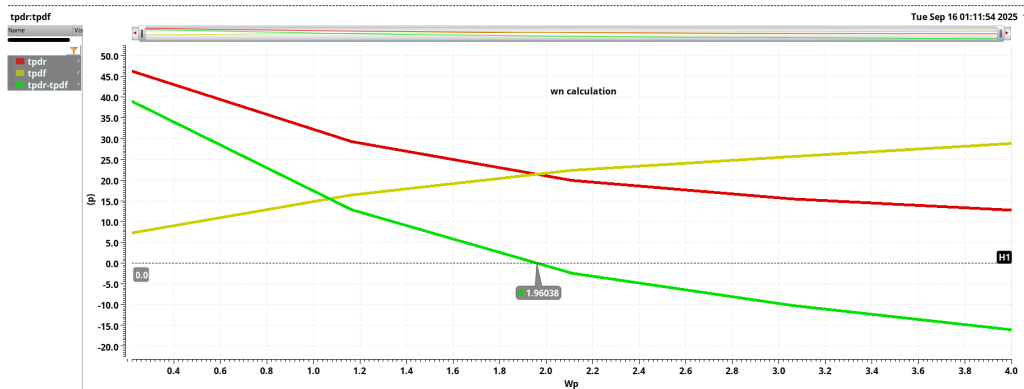
(c) Varying device lengths for fixed Width – 3 values

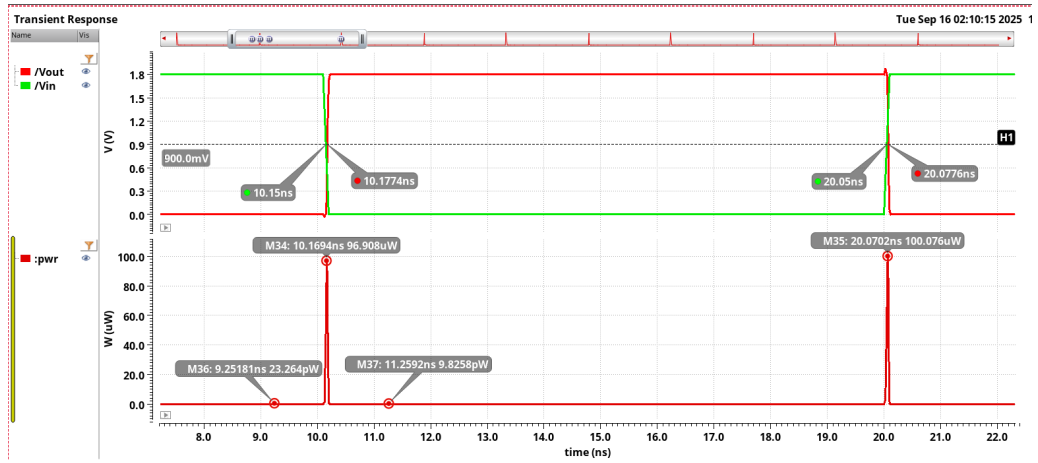
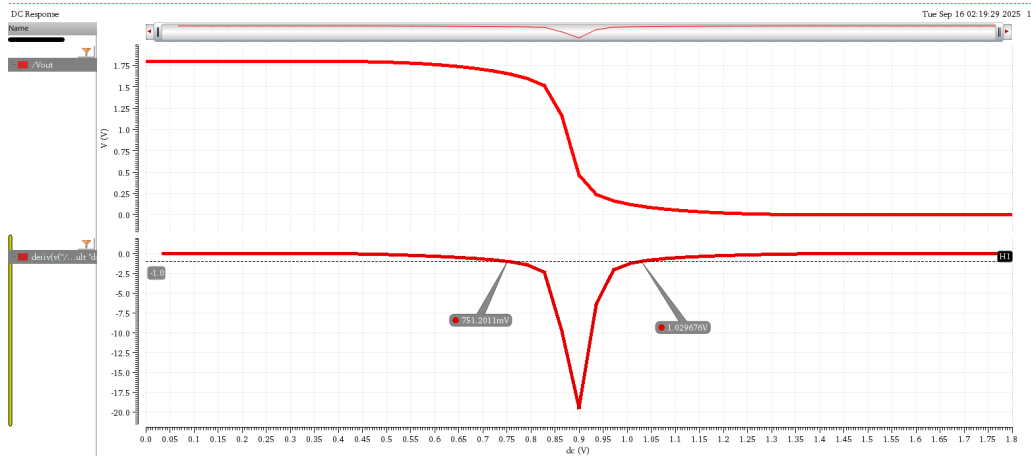
Table 3: Delay, Static Power, Dynamic Power, and Noise Margins for different W_p and L_p values at $V_{DD} = 1.8V$

Parameter	Case 1	Case 2	Case 3
V_{DD} (V)	1.8	1.8	1.8
W_n (nm)	0.22	0.22	0.22
W_p (nm)	0.938293	1.92395	2.23045
L_p (nm)	0.22	0.44	0.66
Rise Delay (ps)	27.43	50.10	67.10
Fall Delay (ps)	27.57	50.30	67.10
P_{stat} $V_{in}=0$ (pW)	9.82	9.83	9.82
P_{stat} $V_{in}=V_{dd}$ (pW)	9.25	14.26	12.48
P_{dyn} LH (μW)	96.90	168.616	250.071
P_{dyn} HL (μW)	100.076	202.29	304.419
NML (mV)	1029.676	1012.08	766.876
NMH (mV)	1048.799	1051.364	1028.825

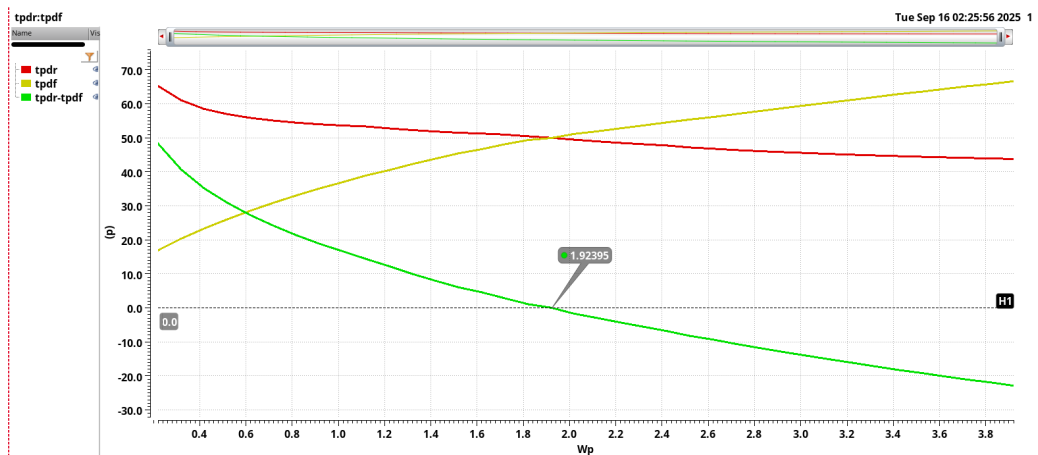
- **Delay:** Both rise and fall delays increase significantly as W_p and L_p increase, due to higher channel resistance.
- **Static Power:** Static power at $V_{in} = 0$ remains nearly constant, while at $V_{in} = V_{DD}$, it increases with larger W_p and L_p , reflecting higher leakage currents.
- **Dynamic Power:** LH and HL dynamic power increase strongly with increasing W_p and L_p , because larger transistors area so charger and discharge increases.
- **Noise Margins:** NML decreases with increasing W_p and L_p after an initial rise, while NMH shows slight improvement.

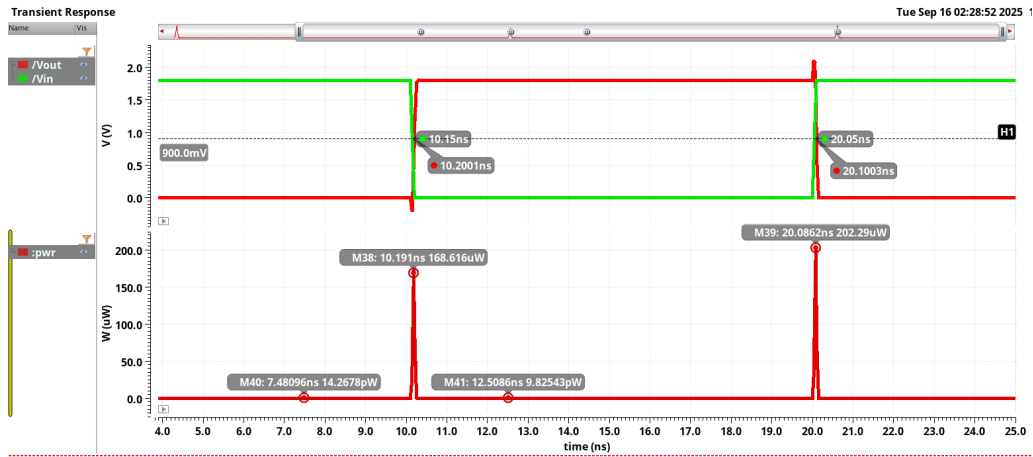
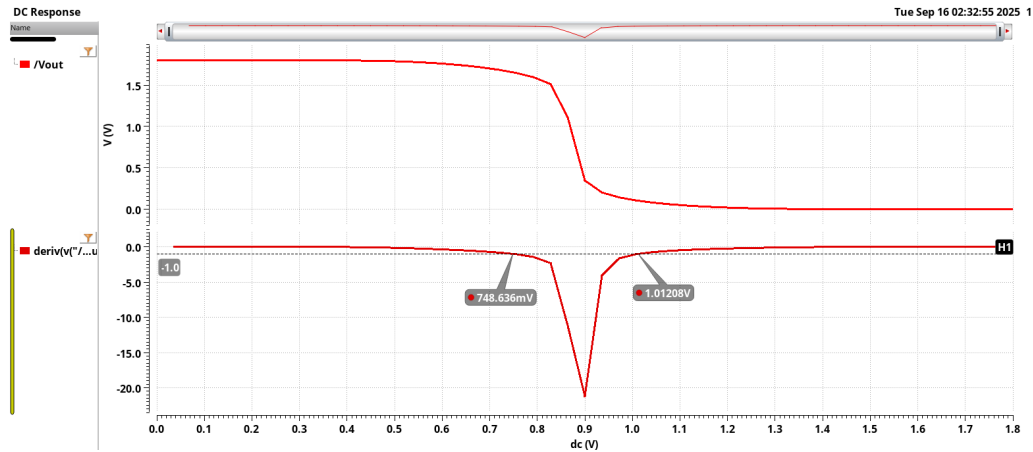
(1) $V_{dd} = 1.8$, $L_n = 0.18nm$, $W_n = 0.22nm$, $L_p = 0.22nm$, $W_p = 0.938293nm$

Figure 14: Calculation of W_p to make delay equal

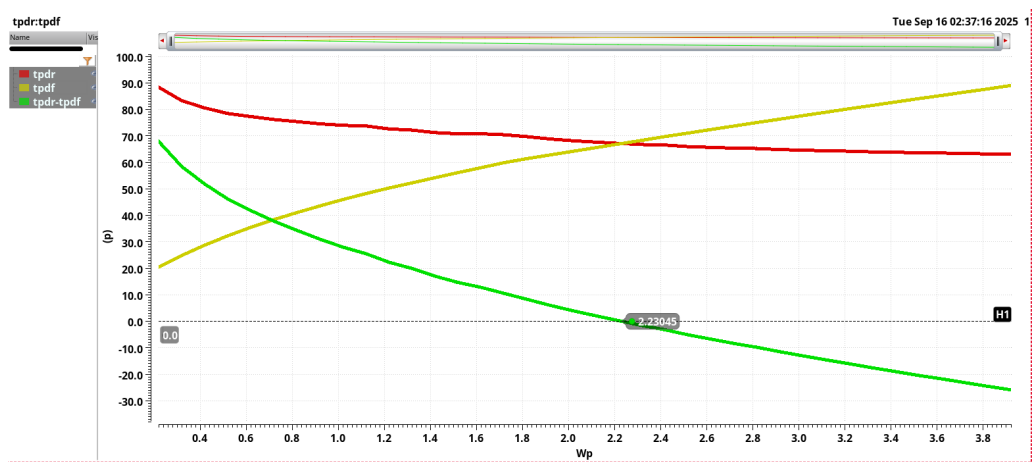
Figure 15: Rise and fall delay and static/dynamic power at $L_p = 0.22\text{nm}$ Figure 16: Noise margin at $L_p = 0.22\text{nm}$

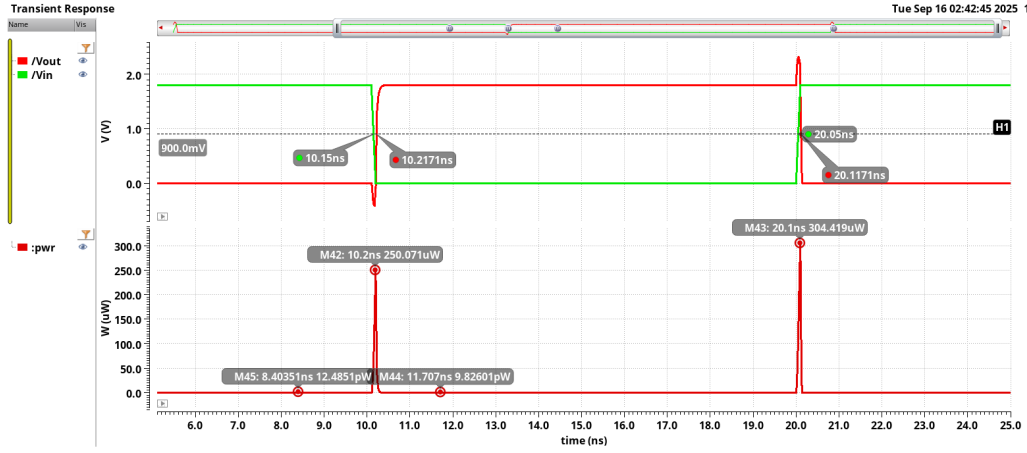
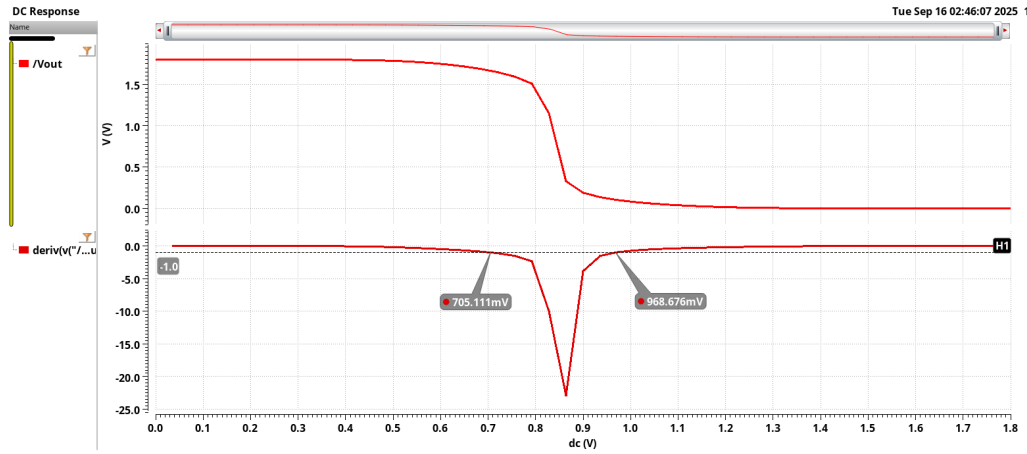
(2) $V_{dd} = 1.8$, $L_n = 0.18\text{nm}$, $W_n = 0.44\text{nm}$, $L_p = 0.44\text{nm}$, $W_p = 1.92395\text{nm}$

Figure 17: Calculation of W_p to make delay equal

Figure 18: Rise and fall delay and static/dynamic power at $L_p = 0.44\text{nm}$ Figure 19: Noise margin at $L_p = 0.44\text{nm}$

(3) $V_{dd} = 1.8$, $L_n = 0.18\text{nm}$, $W_n = 0.66\text{nm}$, $L_p = 0.66\text{nm}$, $W_p = 2.23045\text{nm}$

Figure 20: Calculation of W_p to make delay equal

Figure 21: Rise and fall delay and static/dynamic power at $L_p = 0.66\text{nm}$ Figure 22: Noise margin at $L_p = 0.66\text{nm}$

(d) All the process corners

Table 4: Delay, Static Power, Dynamic Power, and Noise Margins for different Process Corners

Parameter	TT	SS	FF	FS	SF
V_{DD} (V)	1.8	1.8	1.8	1.8	1.5
W_n (nm)	0.22	0.22	0.22	0.22	0.22
W_p (nm)	0.964611	1.24277	0.724367	1.30368	0.567699
Process Corner	TT	SS	FF	FS	SF
Rise Delay (ps)	26.1	32.7	20.0	27.2	23.0
Fall Delay (ps)	26.0	32.8	20.1	27.3	23.1
P_{stat} Vin=0 (pW)	9.82	0.93	0.243	0.049	3.04
P_{stat} Vin=Vdd (pW)	9.20	1.01	0.121	0.20	28.31
P_{dyn} LH (μW)	98.53	98.61	108.342	2.3	76.59
P_{dyn} HL (μW)	95.453	93.0	103.77	107.258	78.91
NML (mV)	1022.68	994.107	1061.15	1009.557	1041.4
NMH (mV)	1043.55	1034.79	1052.18	1054.8752	1037.568

- **Delay:** Maximum at SS (slow devices) and minimum at FF (fast devices), with FS and SF showing in between values.
- **Static Power:** Very low overall; highest leakage observed at the SF corner.
- **Noise Margins:** Remain close to 1 V across all corners, indicating good stability and decreases in immunity noise.

(1) $V_{dd} = 1.8$, $L_n = 0.18\text{nm}$, $W_n = 0.22\text{nm}$, $L_p = 0.18\text{nm}$, $W_p = 0.938293\text{nm}$ for TT

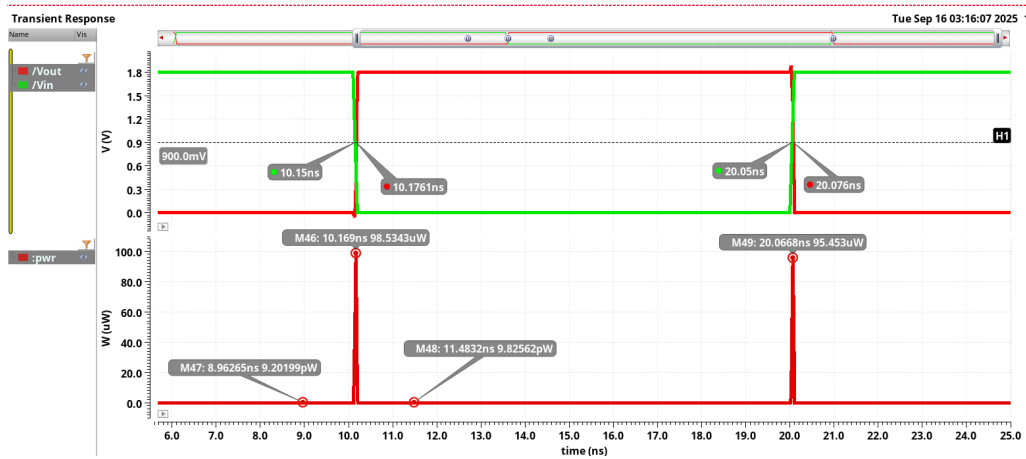


Figure 23: Calculation of W_p to make delay equal

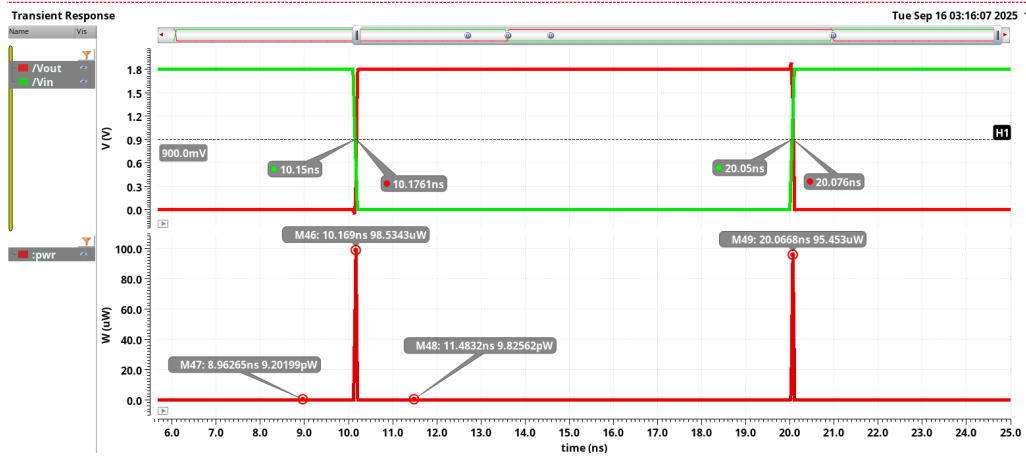


Figure 24: Rise and fall delay and static/dynamic power for TT

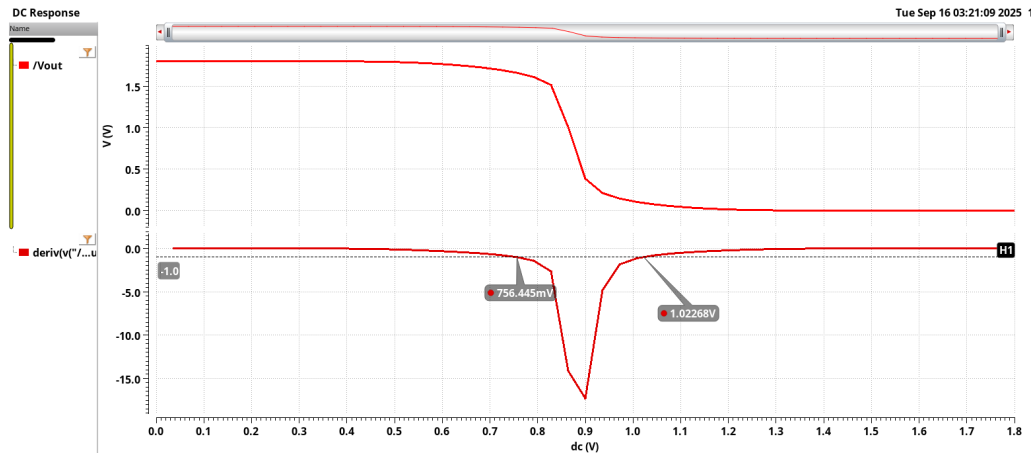


Figure 25: Noise margin for TT

(2) $V_{dd} = 1.8$, $L_n = 0.18\text{nm}$, $W_n = 0.22\text{nm}$, $L_p = 0.18\text{nm}$, $W_p = 1.92395\text{nm}$ for SS

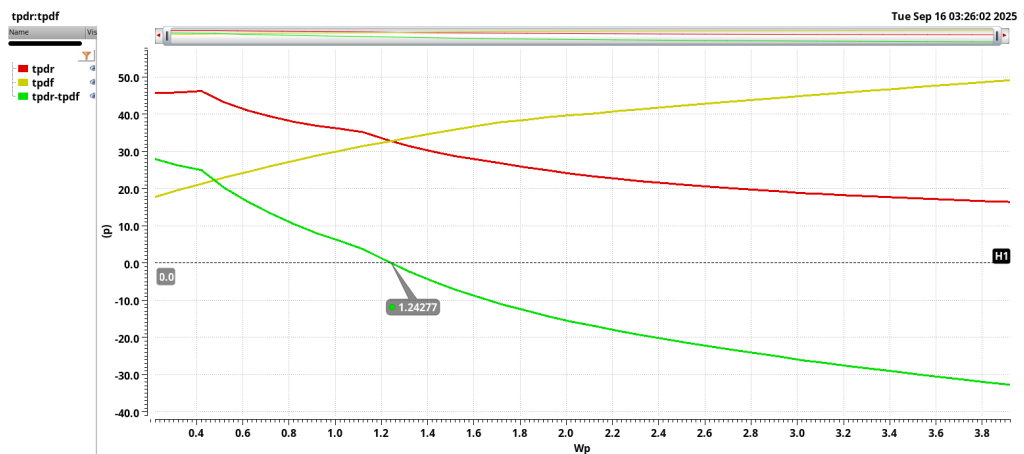
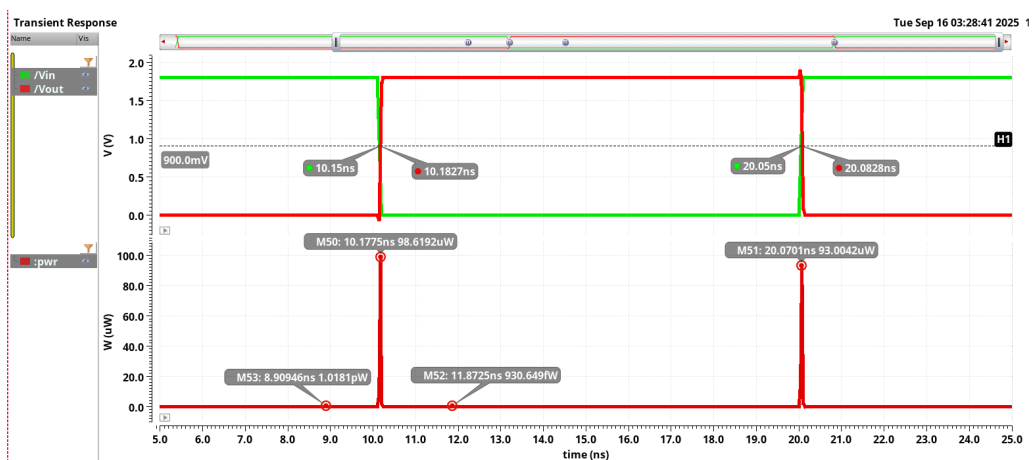
Figure 26: Calculation of W_p to make delay equal

Figure 27: Rise and fall delay and static/dynamic power For SS

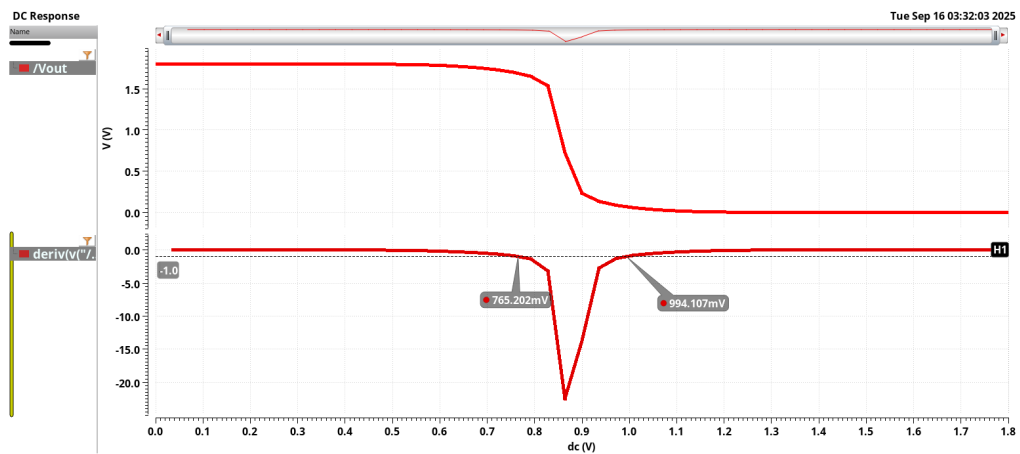


Figure 28: Noise margin For SS

(3) $V_{dd} = 1.8$, $L_n = 0.18\text{nm}$, $W_n = 0.22\text{nm}$, $L_p = 0.18\text{nm}$, $W_p = 2.23045\text{nm}$ for FF

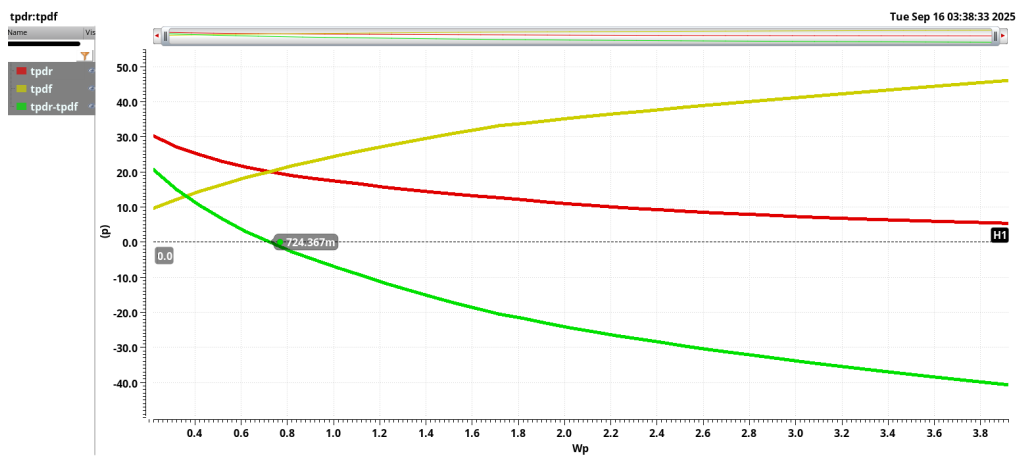


Figure 29: Calculation of Wp to make delay equal

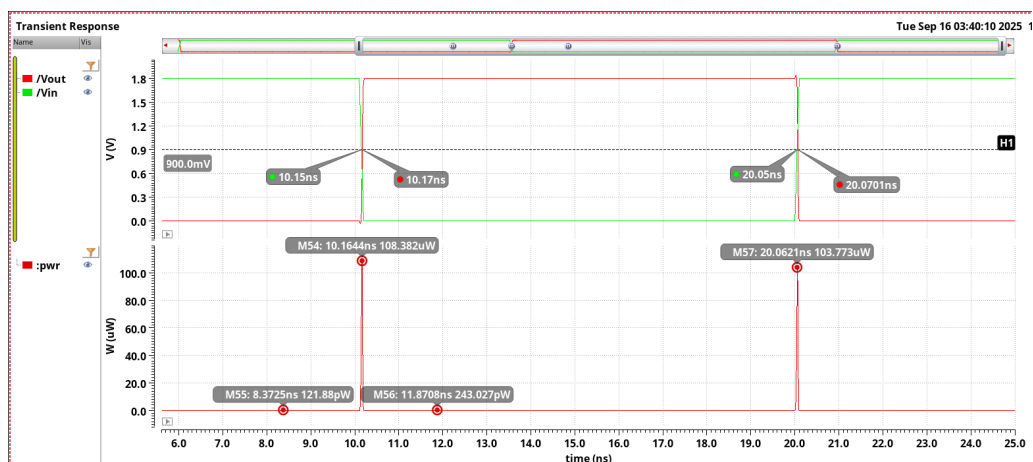


Figure 30: Rise and fall delay and static/dynamic power For FF

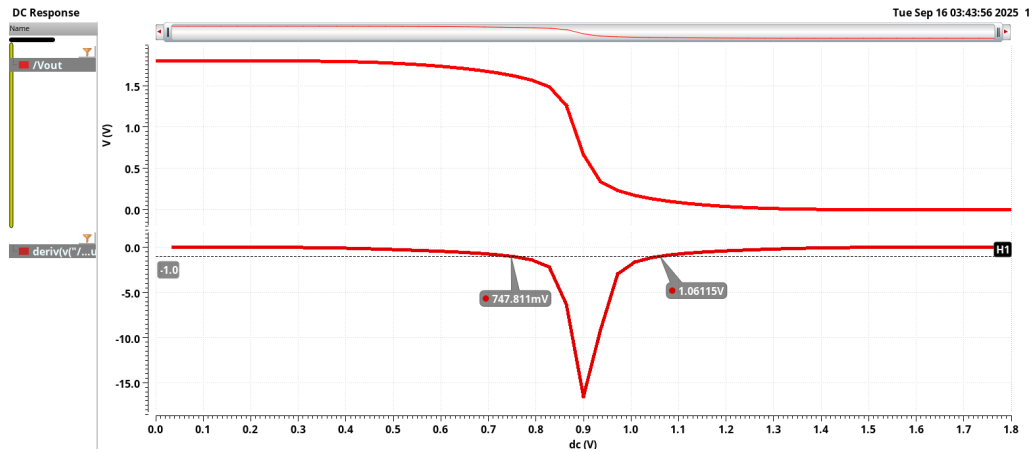


Figure 31: Noise margin For FF

(4) $V_{dd} = 1.8$, $L_n = 0.18\text{nm}$, $W_n = 0.22\text{nm}$, $L_p = 0.18\text{nm}$, $W_p = 1.92395\text{nm}$ for FS

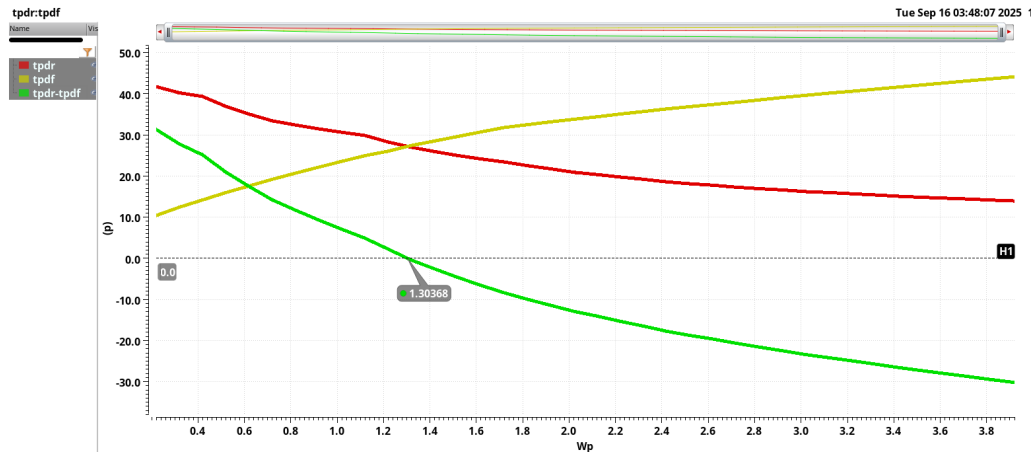
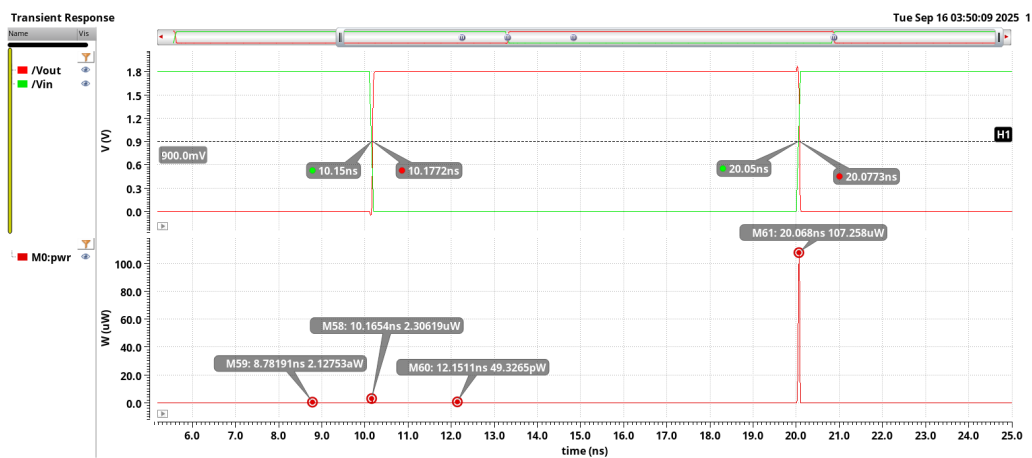
Figure 32: Calculation of W_p to make delay equal

Figure 33: Rise and fall delay and static/dynamic power For FS

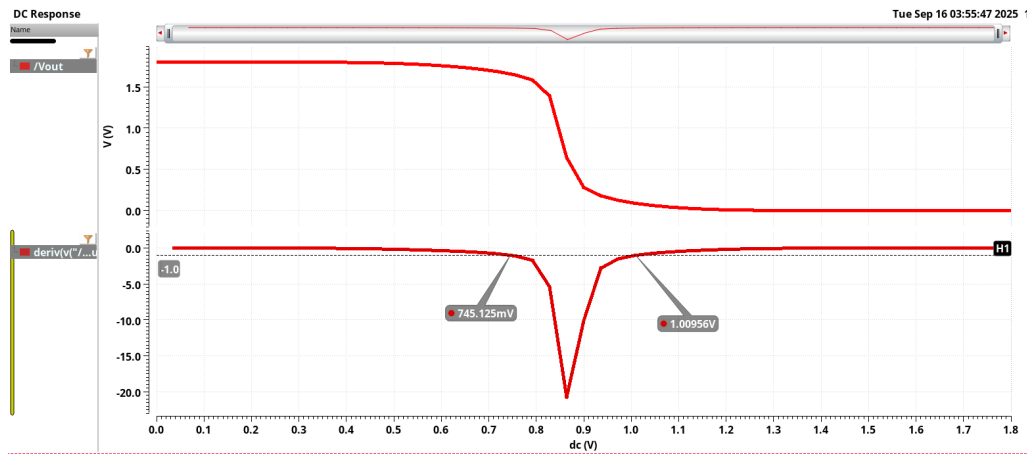


Figure 34: Noise margin For FS

(5) $V_{dd} = 1.8$, $L_n = 0.18\text{nm}$, $W_n = 0.22\text{nm}$, $L_p = 0.18\text{nm}$, $W_p = 2.23045\text{nm}$ for SF

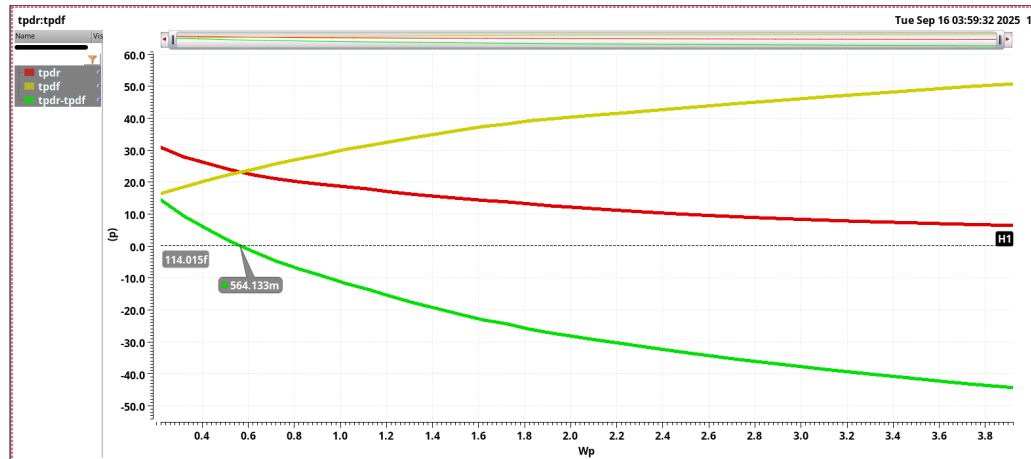
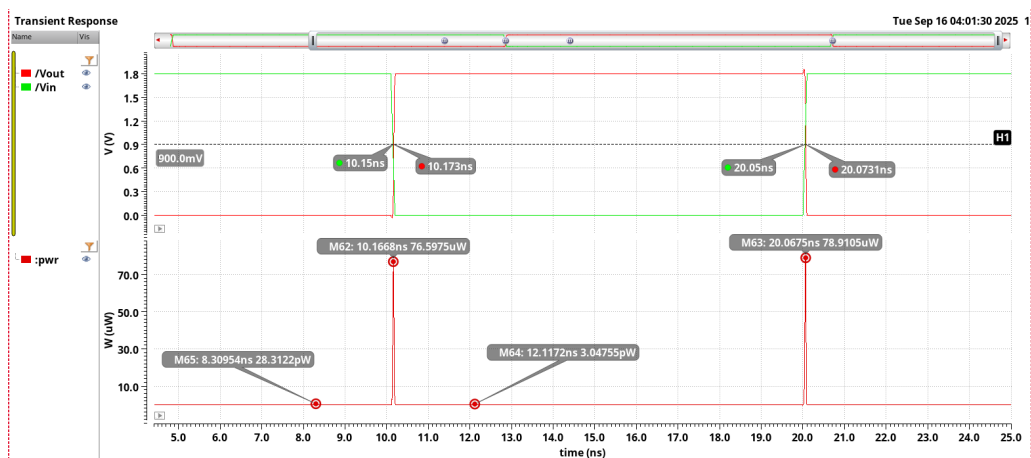
Figure 35: Calculation of W_p to make delay equal

Figure 36: Rise and fall delay and static/dynamic power For SF

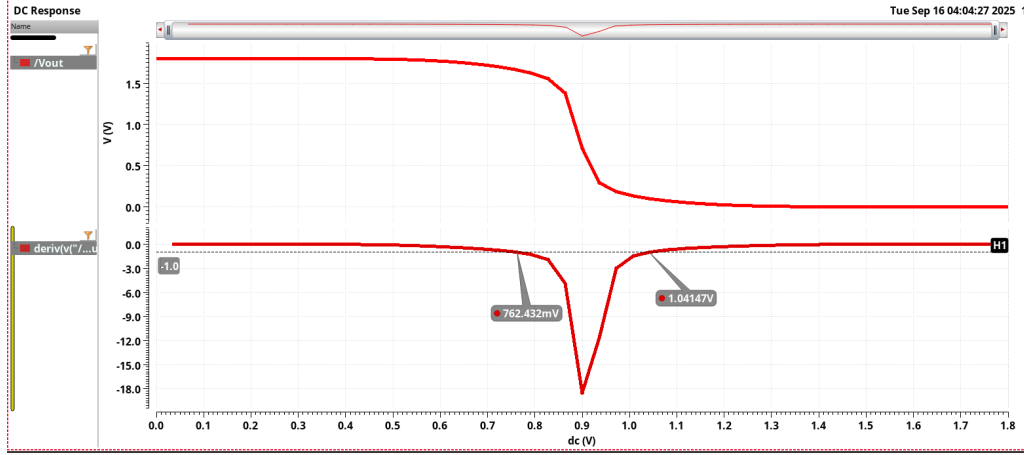


Figure 37: Noise margin For SF

(e) Varying Temperature: -40, 0, 27, 80, 125

Table 5: Delay, Static Power, Dynamic Power, and Noise Margins at different temperatures (W_p , V_{DD} constant)

Parameter	0°C	27°C	40°C	80°C	125°C
V_{DD} (V)	1.8	1.8	1.8	1.8	1.5
W_n (nm)	0.22	0.22	0.22	0.22	0.22
W_p (nm)	1.24277	0.72437	0.96461	1.30368	0.56770
Rise Delay (ps)	32.7	26.1	24.3	27.1	27.8
Fall Delay (ps)	32.8	26.0	24.4	27.0	27.9
P_{stat} Vin=0 (pW)	1.43	9.82	36.14	180.27	113.7
P_{stat} Vin= V_{DD} (fW)	1.57	9.20	54.44	133.31	730.79
P_{dyn} LH (μ W)	99.63	98.53	106.43	92.12	87.48
P_{dyn} HL (μ W)	97.79	95.45	100.68	90.50	86.16
NML (mV)	1013.6	1022.7	1003.0	1037.5	1049.1
NMH (mV)	1040.3	1043.6	1034.7	1053.2	1060.4

- **Delay:** Both rise and fall delays increase significantly with larger W_p and L_p , due to increased channel resistance and capacitance.
- **Static Power:** Static power at $V_{in} = 0$ remains almost constant, while at $V_{in} = V_{DD}$, it increases slightly with larger W_p and L_p , reflecting higher leakage.
- **Dynamic Power:** LH and HL dynamic power increase strongly as W_p and L_p increase, because area increase for charging and discharging increases more capacitance.
- **Noise Margins:** NML first increases and then decreases with larger sizes, while NMH shows a slight increase.

(1) $V_{dd} = 1.8$, $L_n = 0.18\text{nm}$, $W_n = 0.22\text{nm}$, $L_p = 0.18\text{nm}$, $W_p = 0.938293\text{nm}$ for Temp = -40

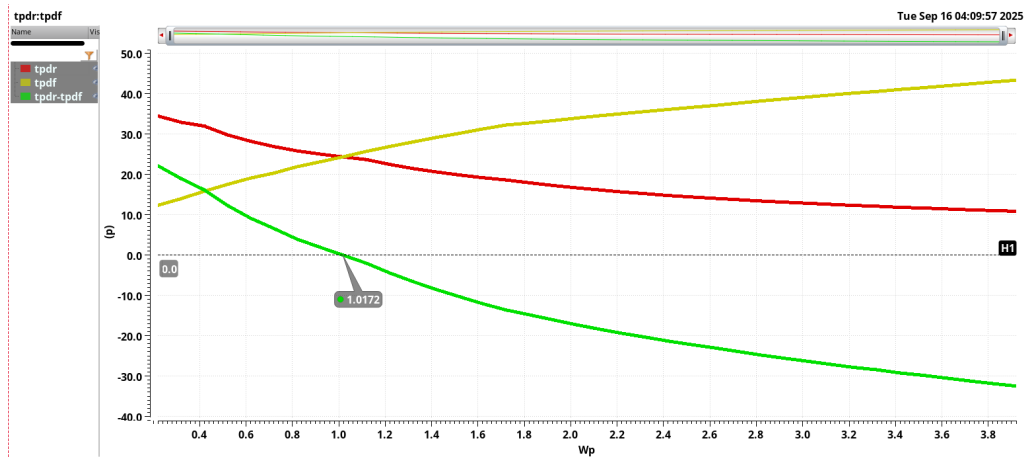


Figure 38: Calculation of Wp to make delay equal at Temp = -40°C

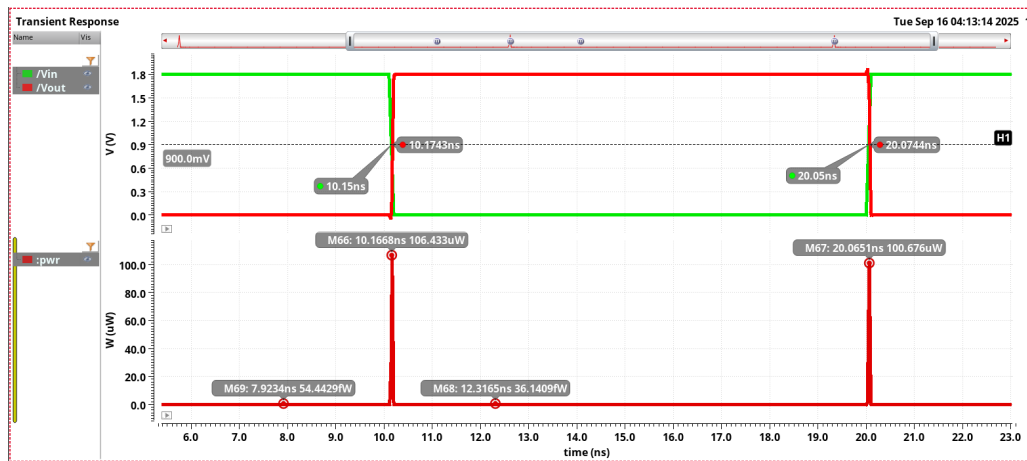


Figure 39: Rise and fall delay and static/dynamic power at Temp = -40°C

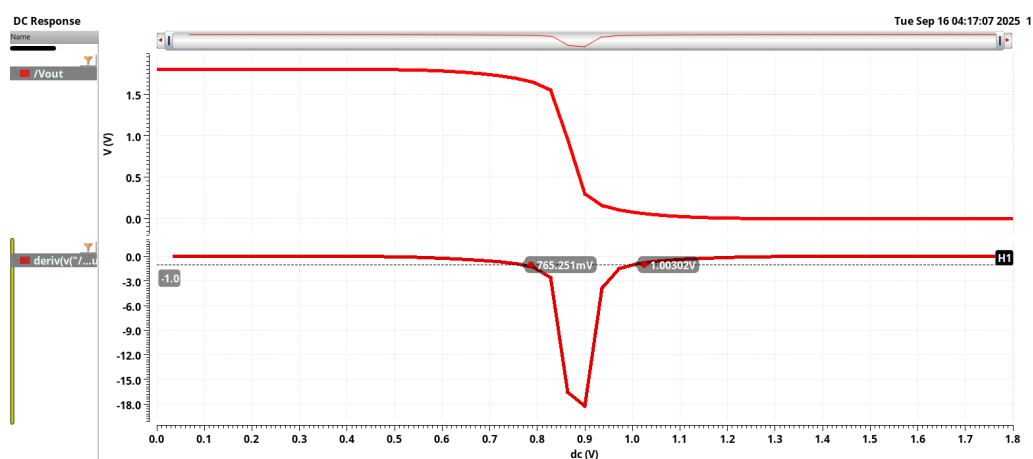


Figure 40: Noise margin at Temp = -40°C

(2) $V_{dd} = 1.8$, $L_n = 0.18\text{nm}$, $W_n = 0.22\text{nm}$, $L_p = 0.18\text{nm}$, $W_p = 1.92395\text{nm}$ for Temp = 0

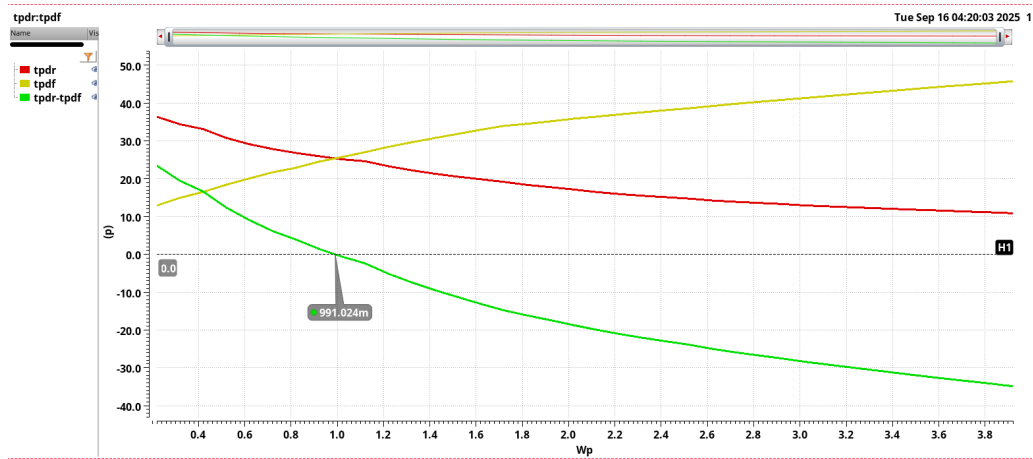


Figure 41: Calculation of Wp to make delay equal at Temp = 0°C

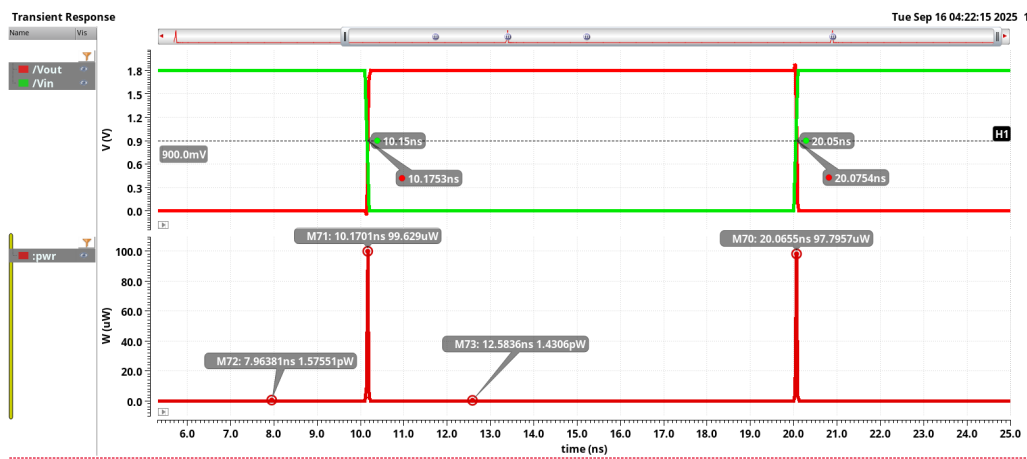


Figure 42: Rise and fall delay and static/dynamic power at Temp = 0°C

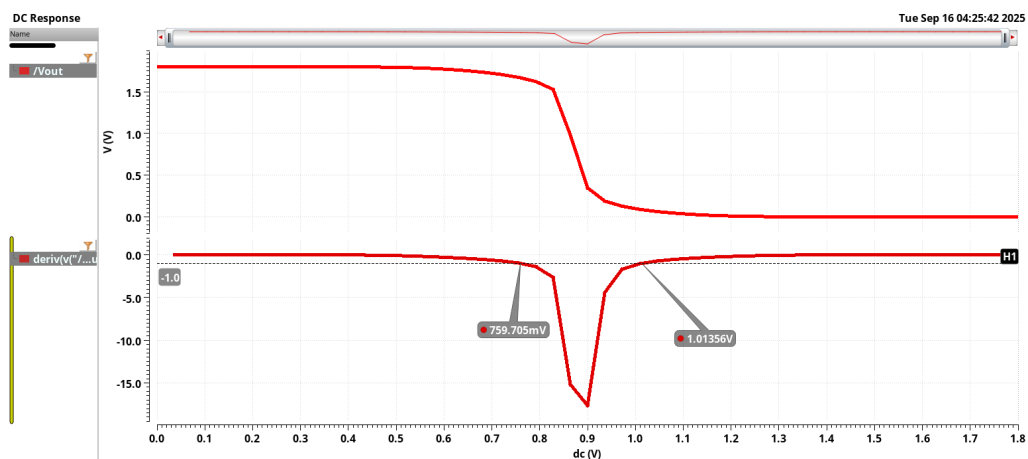


Figure 43: Noise margin at Temp = 0°C

(3) $V_{dd} = 1.8$, $L_n = 0.18\text{nm}$, $W_n = 0.22\text{nm}$, $L_p = 0.18\text{nm}$, $W_p = 2.23045\text{nm}$ for Temp = 27

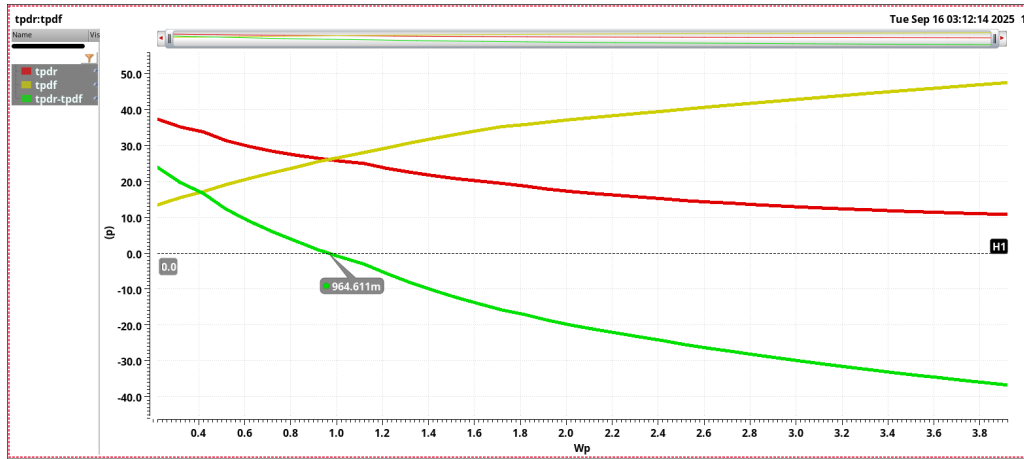


Figure 44: Calculation of Wp to make delay equal at Temp = 27°C

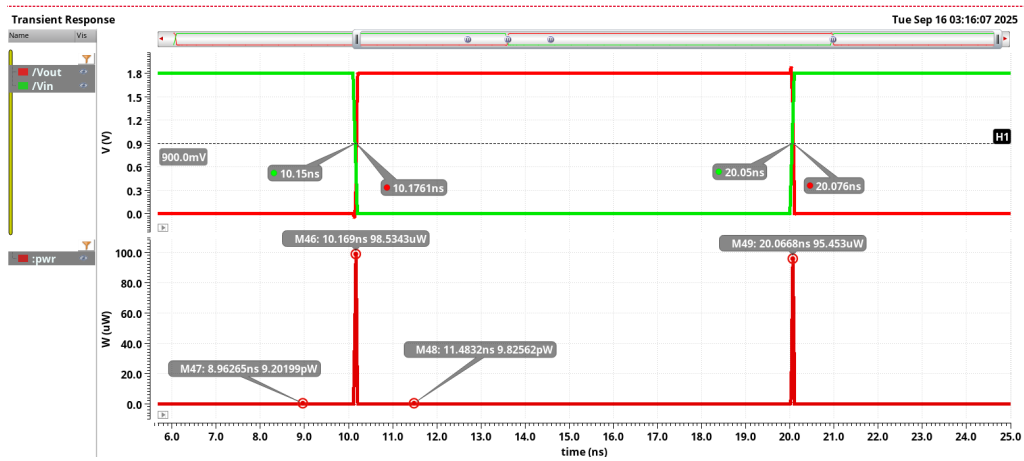


Figure 45: Rise and fall delay and static/dynamic power at Temp = 27°C

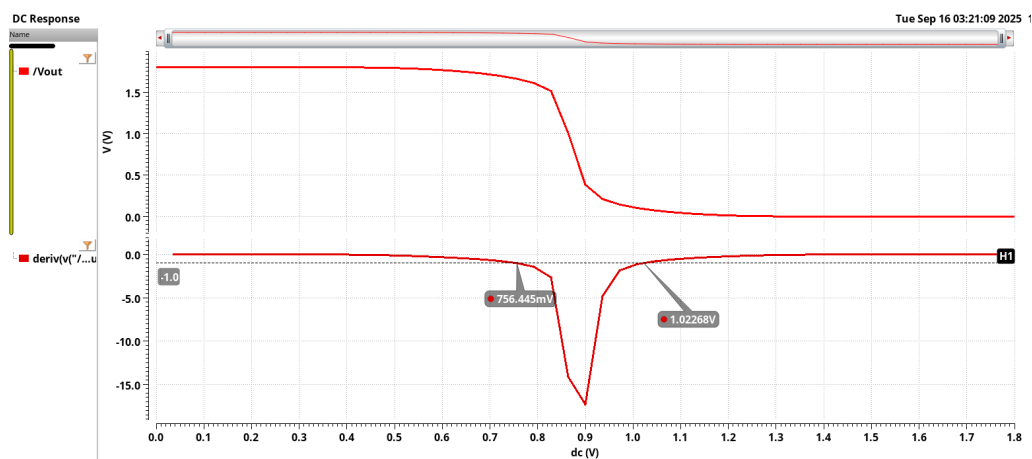


Figure 46: Noise margin at Temp = 27°C

(4) $V_{dd} = 1.8$, $L_n = 0.18\text{nm}$, $W_n = 0.22\text{nm}$, $L_p = 0.18\text{nm}$, $W_p = 1.92395\text{nm}$ for Temp = 80

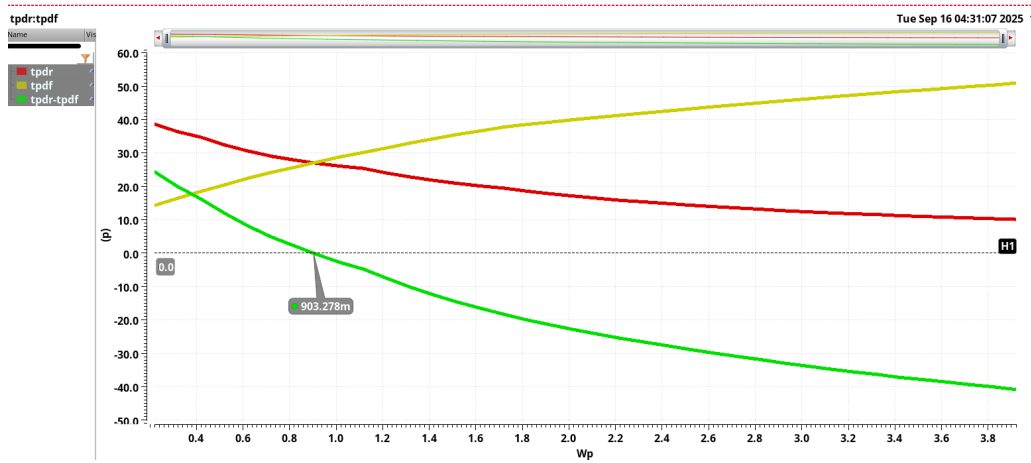


Figure 47: Calculation of Wp to make delay equal at Temp = 80°C

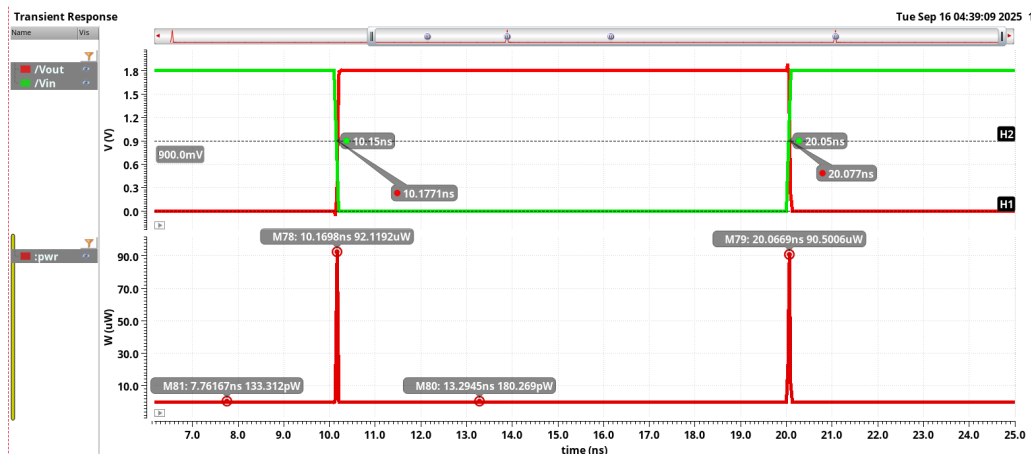


Figure 48: Rise and fall delay and static/dynamic power at Temp = 80°C

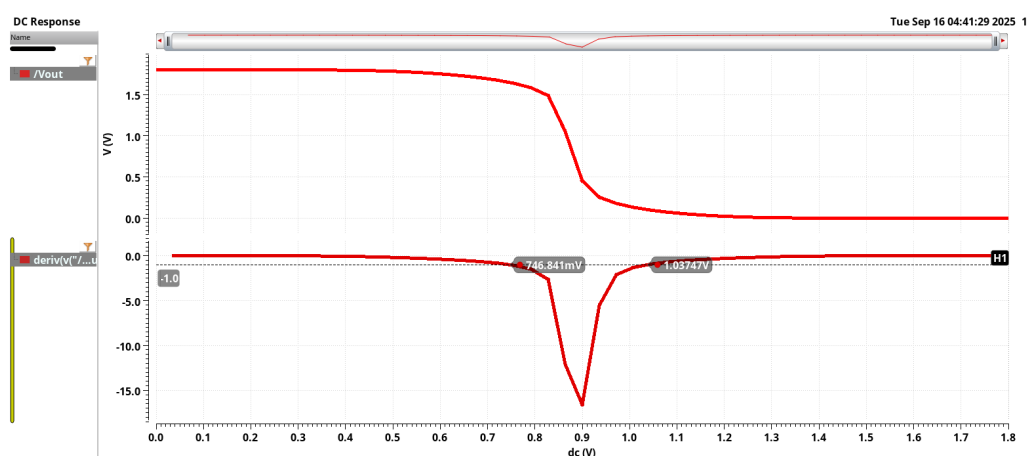


Figure 49: Noise margin at Temp = 80°C

(5) $V_{dd} = 1.8$, $L_n = 0.18\text{nm}$, $W_n = 0.22\text{nm}$, $L_p = 0.18\text{nm}$, $W_p = 2.23045\text{nm}$ for Temp = 125

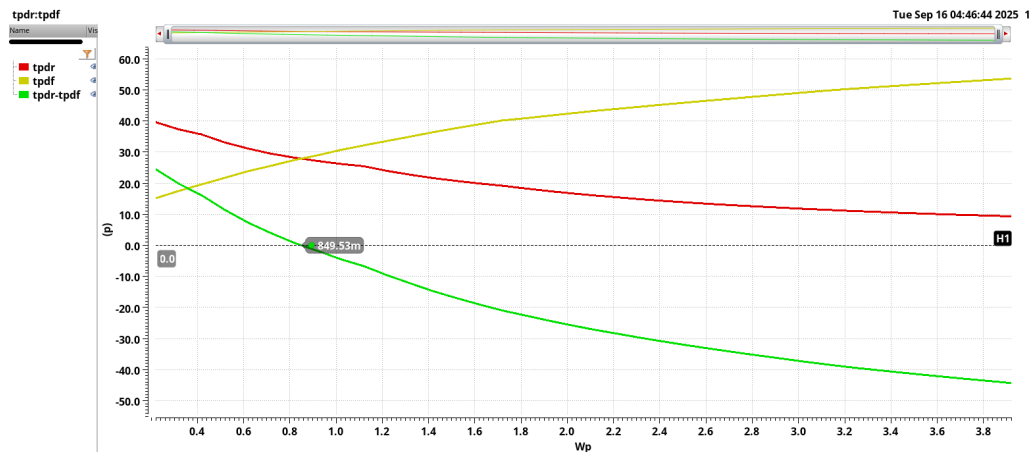
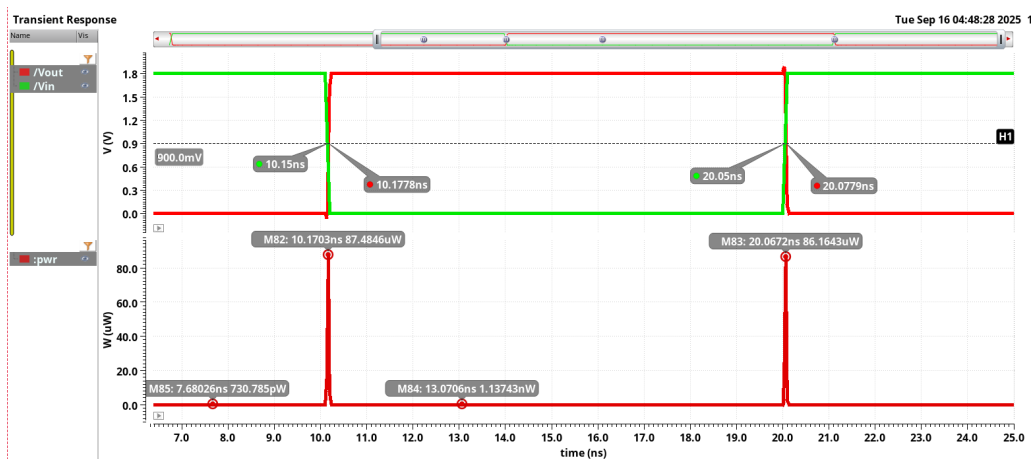
Figure 50: Calculation of W_p to make delay equal at Temp = 125°C

Figure 51: Rise and fall delay and static/dynamic power at Temp = 125°C

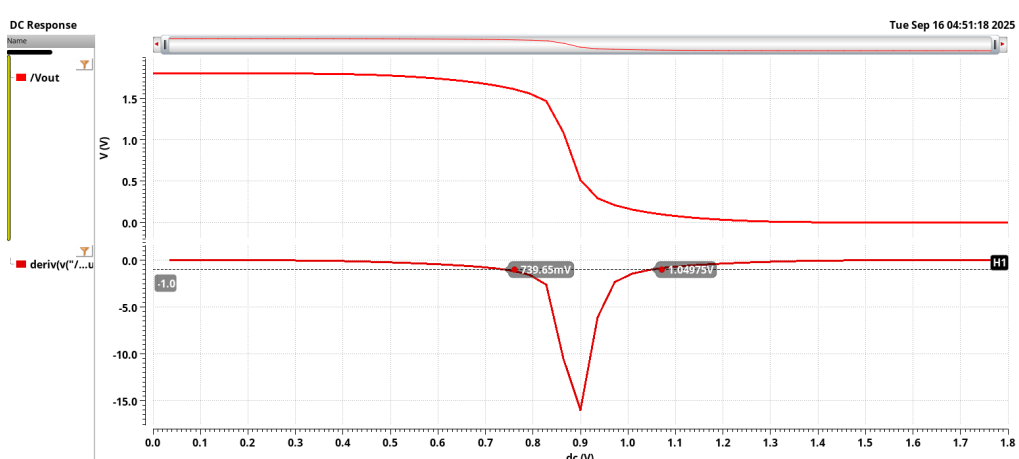


Figure 52: Noise margin at Temp = 125°C

Question 2

Design an inverter with $I_{on_pmos}/I_{on_nmos} = 0.5, 1, 2, 4, 8$ with 10 fF load, and show all the VTCs on a single plot. Find the NML and NMH for all the cases. Explain the results. Also, report the delay, rise time, fall time, static power, and dynamic energy in a table.

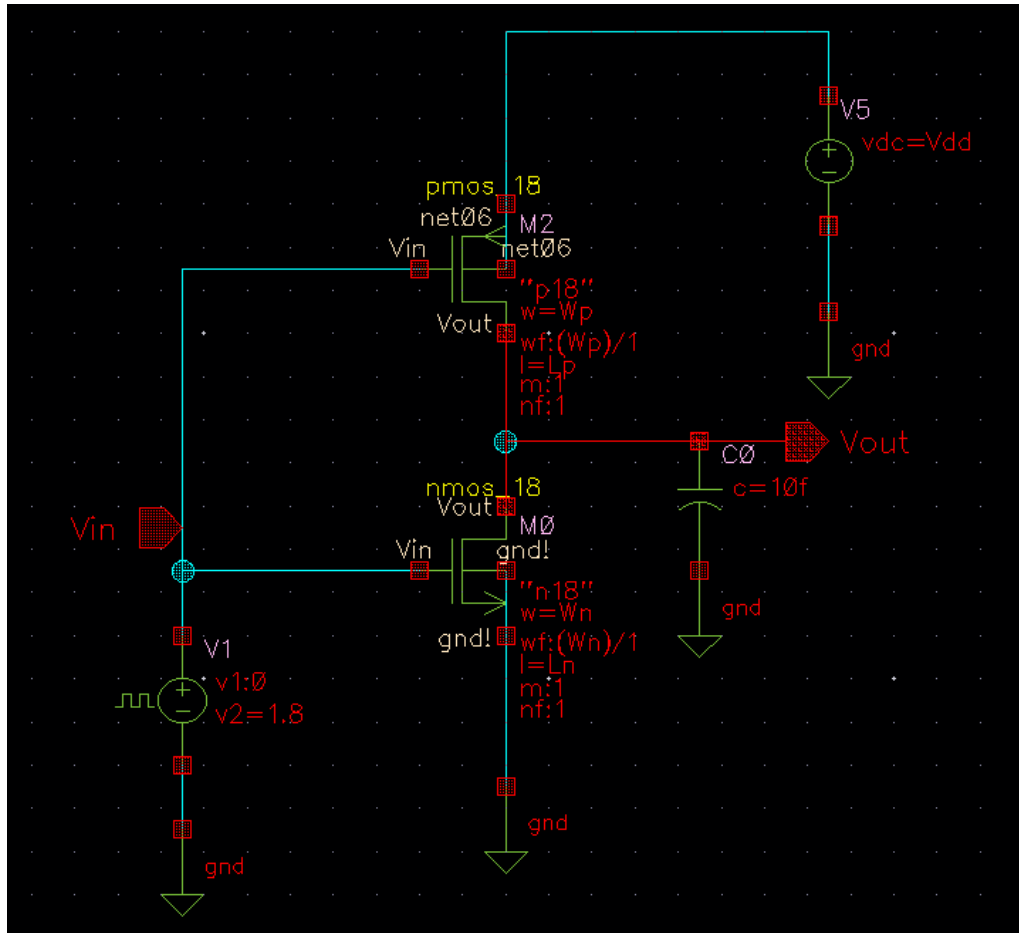


Figure 53: Schematic of Inverter with 20fF capacitor load

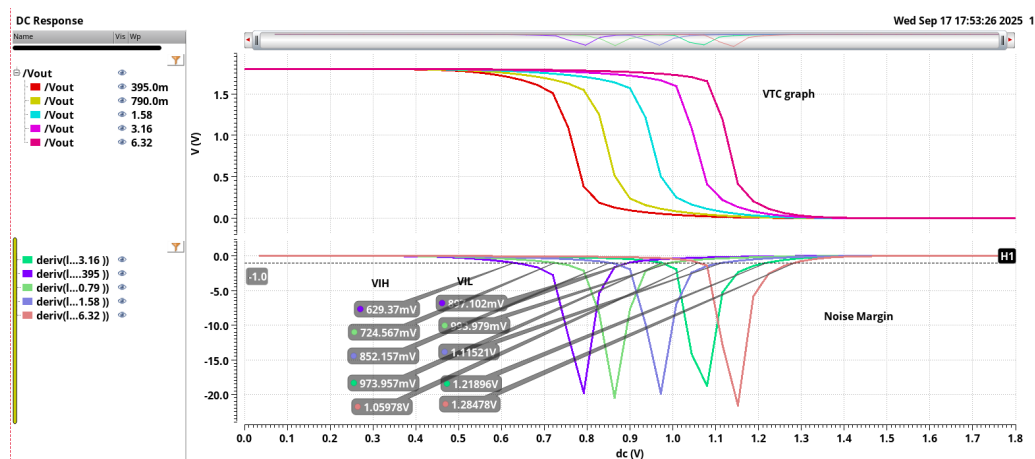


Figure 54: Noise margin and VTC curve for different I_{on_pmos} and I_{on_nmos} ratio

Parameter	1	2	3	4	5
Vdd (V)	1.8	1.8	1.8	1.8	1.8
$I_{on_{pmos}}/I_{on_{nmos}}$	0.5	1	2	4	8
Wp (nm)	0.395	0.789877	1.58	3.16	6.32
Delay rise (ps)	148.73	94.61	58.33	37.88	25.05
Delay fall (ps)	90.12	94.44	102.15	110.22	123.71
Static Power Vin=0 (pW)	9.82	9.82	13.8	13.06	13.15
Static Power Vin=Vdd (pW)	2.455	6.97	18.58	7.04	95.78
Dynamic Power LH (μ W)	134.7	228.22	358.59	537.95	10.165
Dynamic Power HL (μ W)	204.74	211.49	221.49	232.97	249.355
NML = Vil (mV)	897.1024	993.957	1115.2	1218.95	1284.78
NMH = Vdd - Vih (mV)	1170.63	1075.453	947.84	826.04	740.23

Table 6: Vertical representation of Delay, Static Power, Dynamic Power, and Noise Margins for different $I_{on_{pmos}}/I_{on_{nmos}}$ ratios

- **Delay:** Rise delay decreases and fall delay increases as $I_{on_{pmos}}/I_{on_{nmos}}$ ratio increases, showing faster pull-up and slower pull-down transitions with stronger PMOS.
- **Static Power:** Static power at $V_{in} = 0$ slightly increases with higher ratios, while at $V_{in} = V_{DD}$, it increases sharply at very high ratios due to stronger PMOS leakage.
- **Noise margin:** As the PMOS/NMOS current ratio increases, NML increases, NMH decreases, and the effective noise margin (minimum of the two) becomes worse.

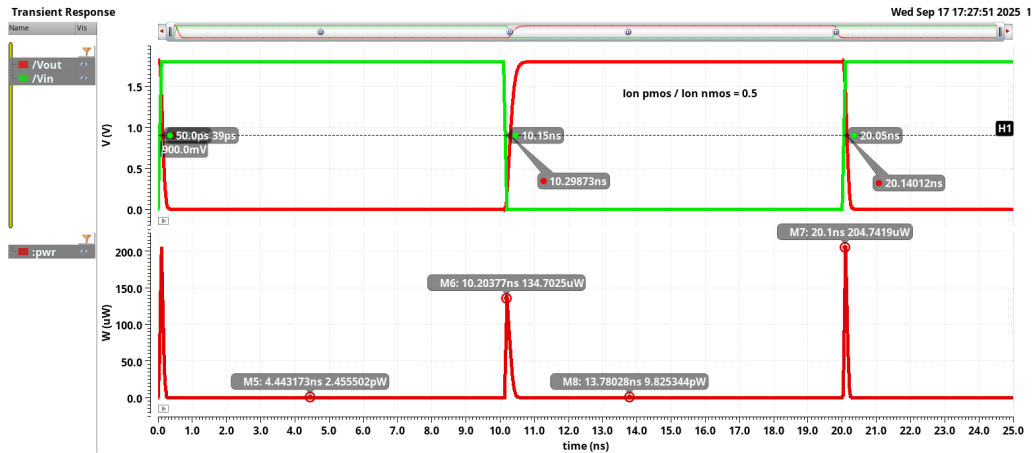


Figure 55: Rise and fall delay and static/dynamic power at $I_{onPmos}/I_{onNmos} = 0.5$

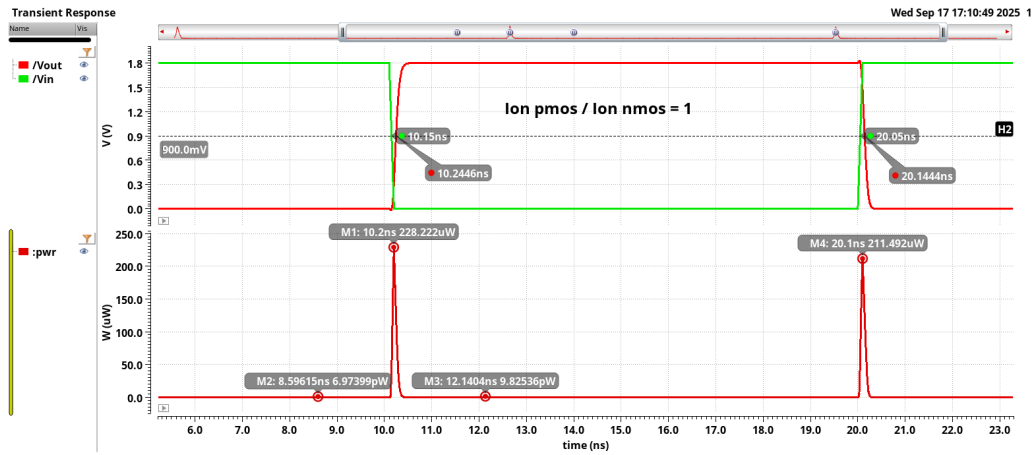


Figure 56: Rise and fall delay and static/dynamic power at IonPmos/IonNmos = 1

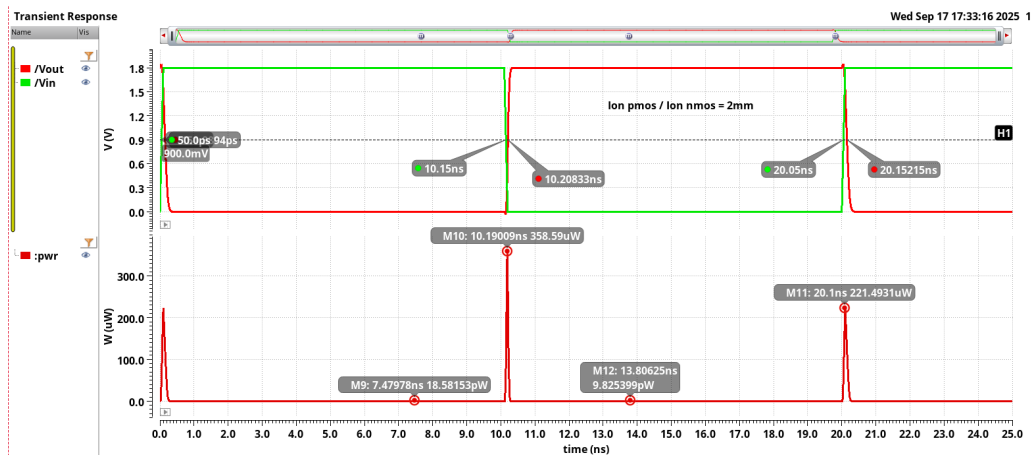


Figure 57: Rise and fall delay and static/dynamic power at IonPmos/IonNmos = 2

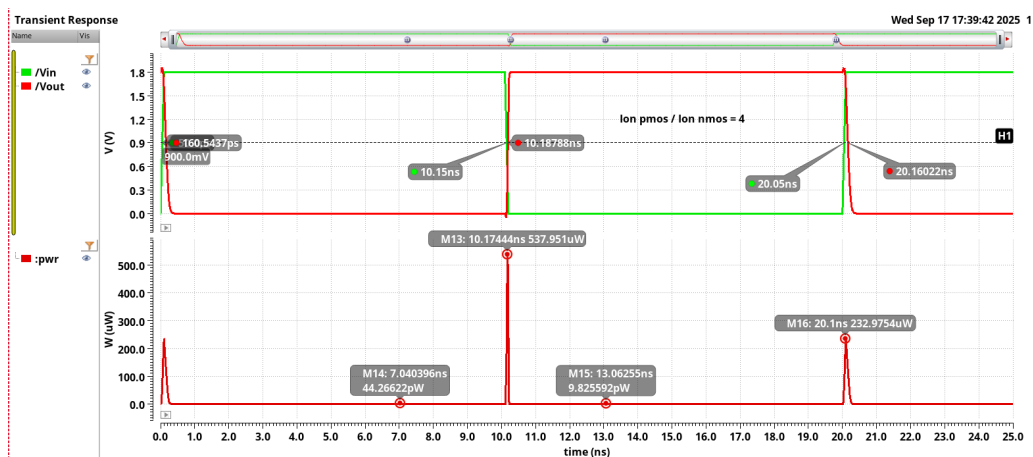


Figure 58: Rise and fall delay and static/dynamic power at IonPmos/IonNmos = 4

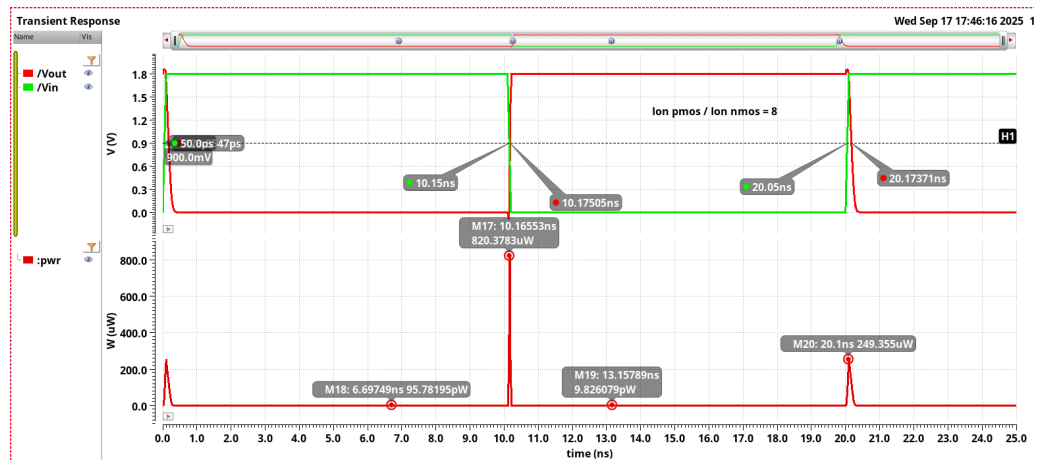


Figure 59: Rise and fall delay and static/dynamic power at IonPmos/IonNmos = 8

Question 3

Design a five-stage ring oscillator. Report frequency for the following cases.

FF(125°C), FF(-40°C), SS(125°C), SS(-40°C), TT(25 °C)

Explain your observations.

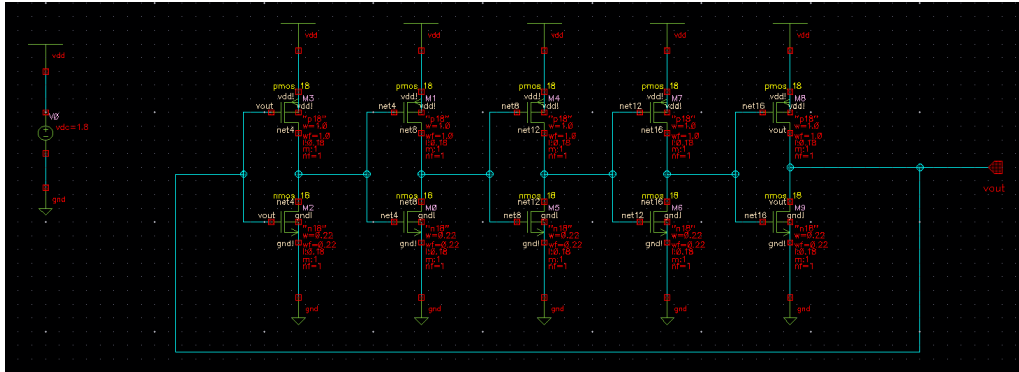


Figure 60: Ring Oscillator

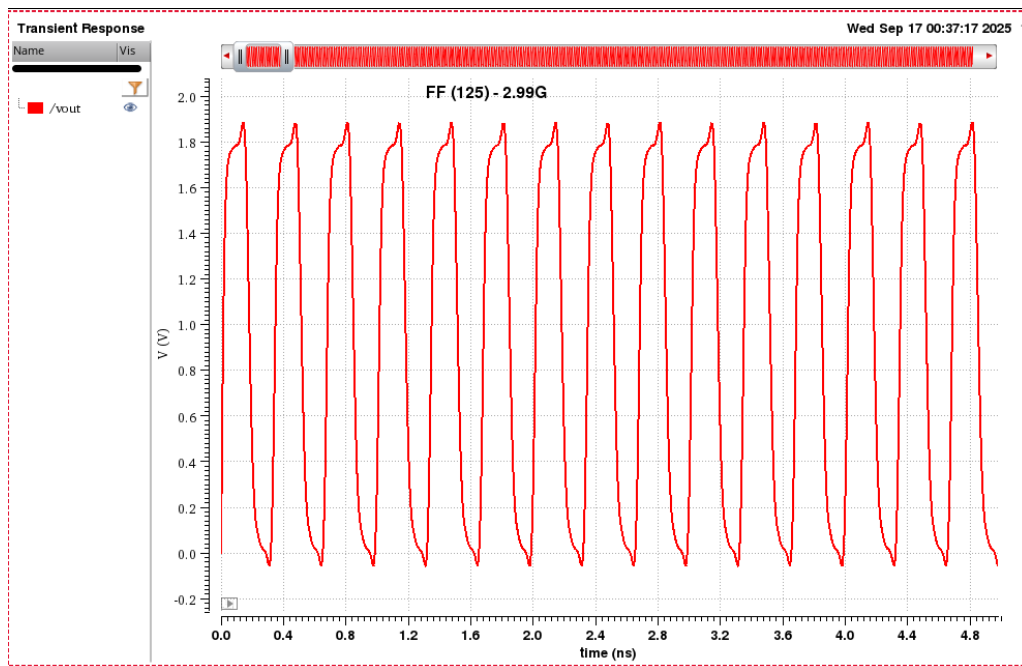


Figure 61: Frequency Waveform for FF(125°C)

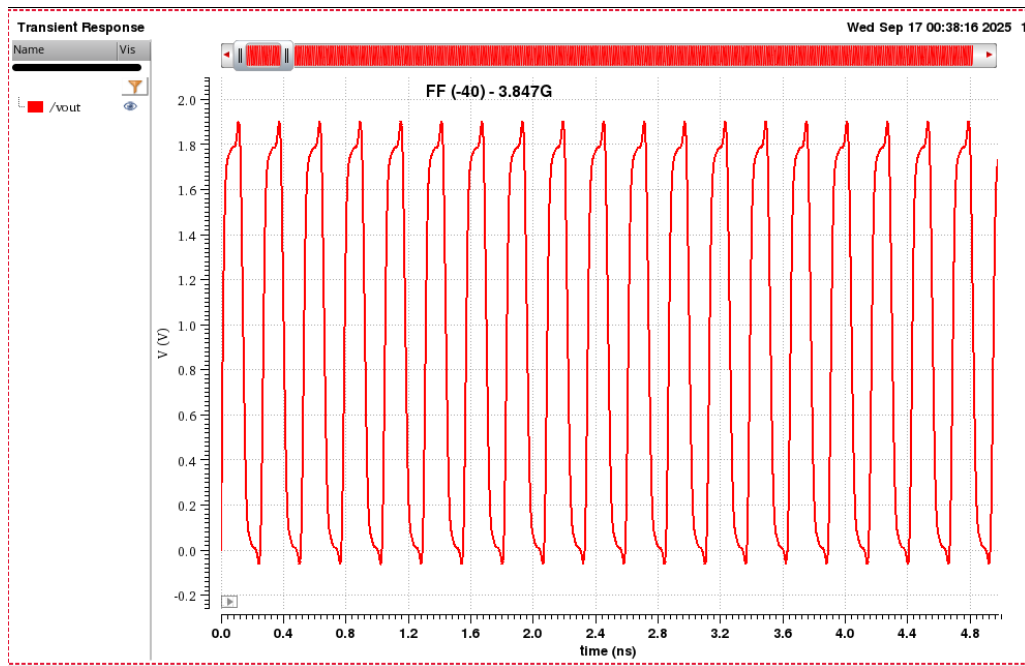


Figure 62: Frequency Waveform for FF(-40°C)

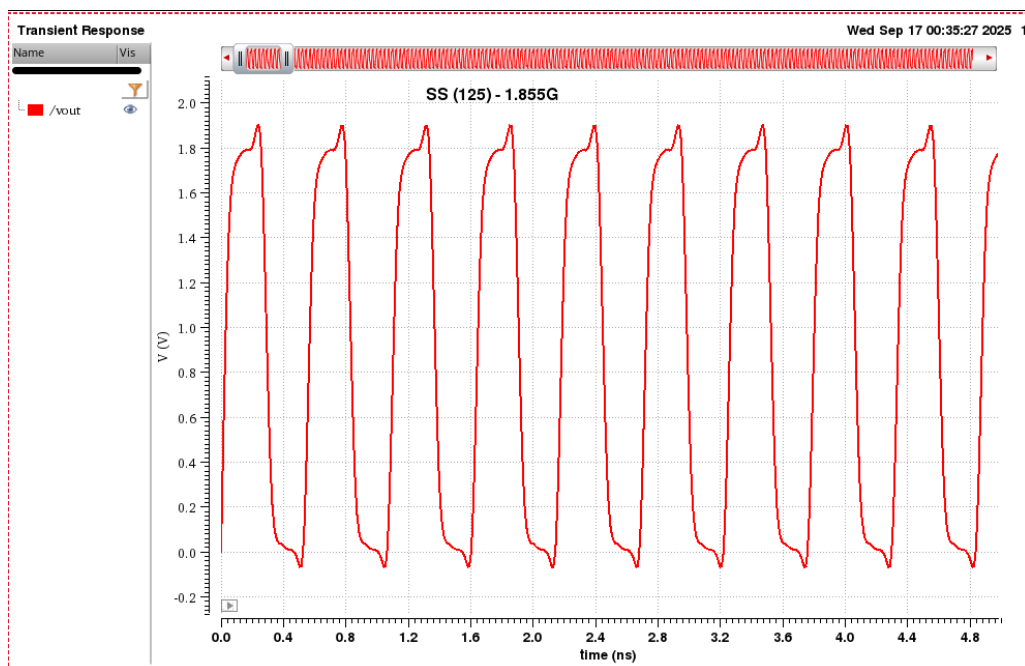


Figure 63: Frequency Waveform for SS(125°C)

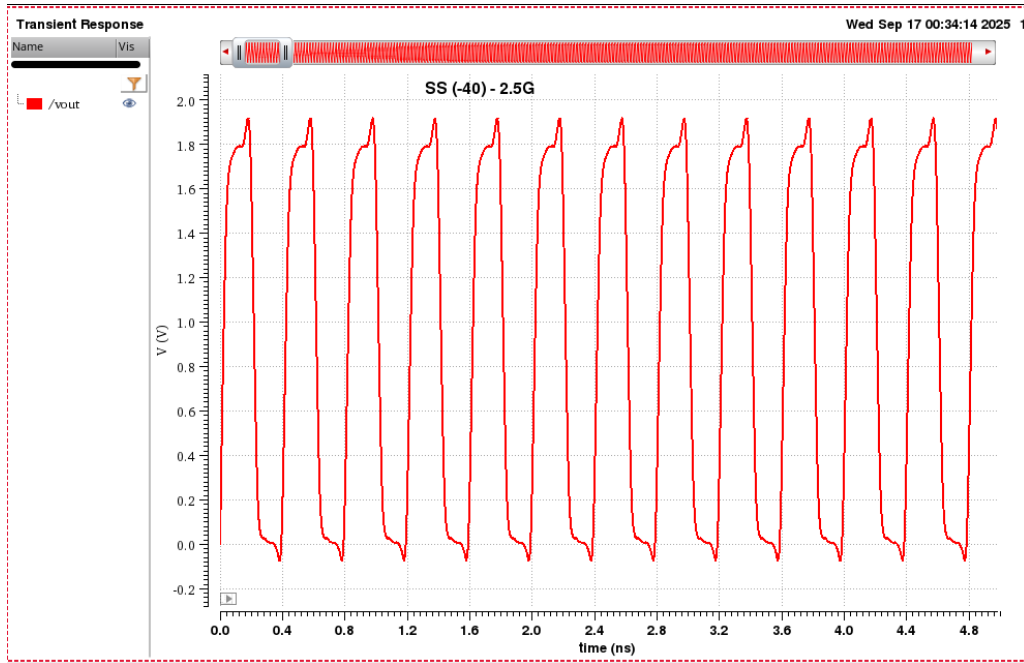


Figure 64: Frequency Waveform for SS(-40°C)

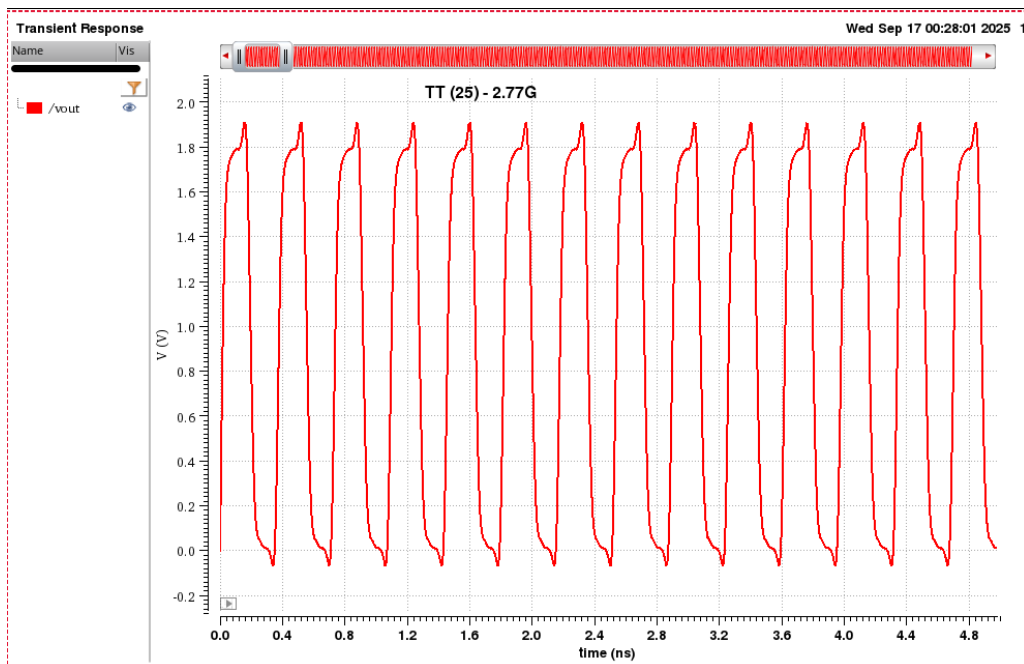


Figure 65: Frequency Waveform for TT(25°C)

Observations

- The Frequency trend is

$$FF(-40^{\circ}\text{C}) > FF(125^{\circ}\text{C}) > TT(25^{\circ}\text{C}) > SS(-40^{\circ}\text{C}) > SS(125^{\circ}\text{C})$$

Question 4

4. [Marks: 40] Implement the following gates using static CMOS logic. Size all the transistors such that the rise and fall delays are equal. Verify the Truth table and report the total dynamic energy and delay. Explain the result you have obtained.

The functions are the following:

- (a) NAND2 (2-input NAND)

Table 7: Delay and Dynamic Energy Results

Parameter	Value
Rise Delay (ps)	69.75
Fall Delay (ps)	70.53
Dynamic Energy	—

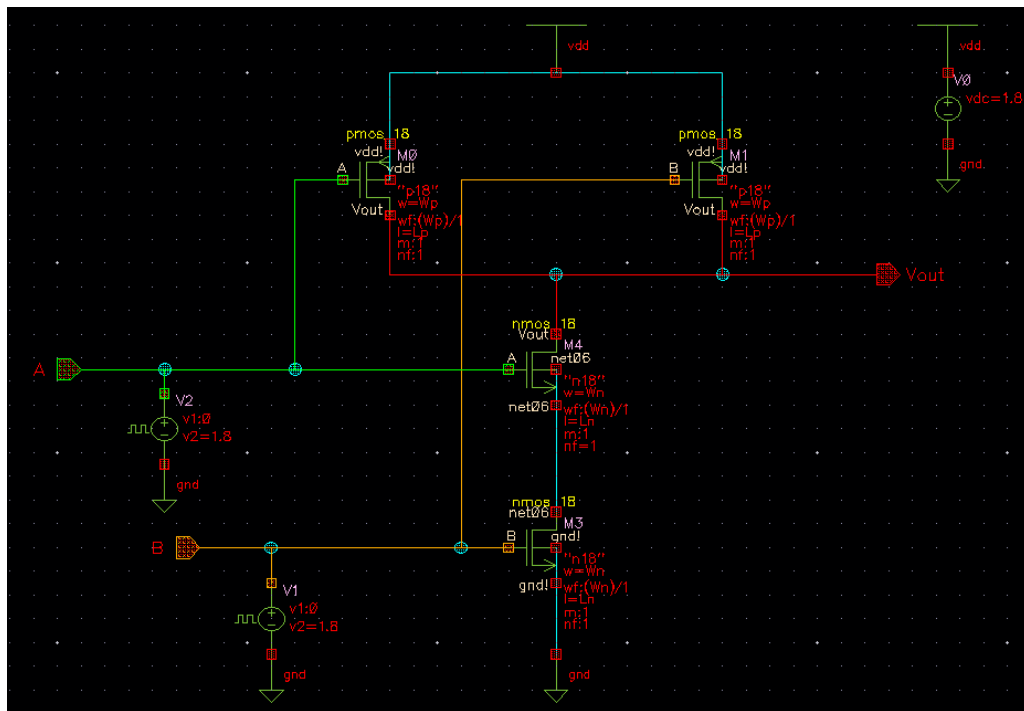


Figure 66: Schematic of NAND Gate

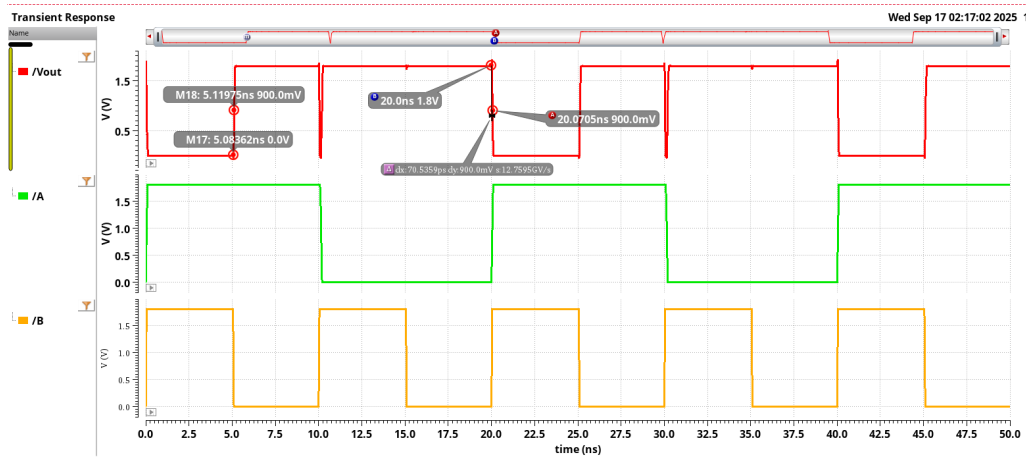


Figure 67: Rise and fall Delay

(b) NOR2 (2-input NOR)

Table 8: Delay and Dynamic Energy Results

Parameter	Value
Rise Delay (ps)	37.809
Fall Delay (ps)	41.2
Dynamic Energy	3.93E-10

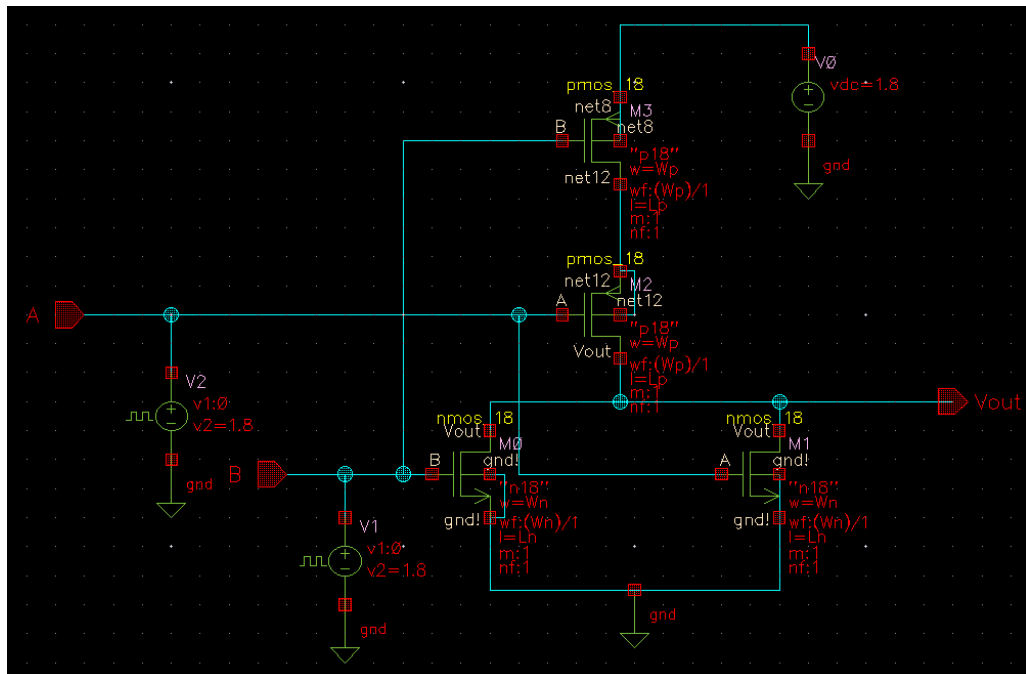


Figure 68: Schematic of NOR Gate

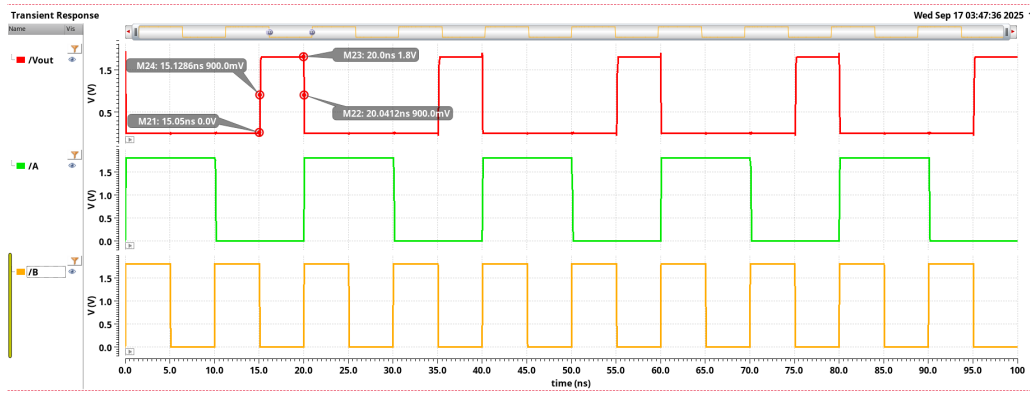


Figure 69: Rise and fall Delay

(c) XNOR2 (2-input XNOR)

Table 9: Delay and Dynamic Energy Results

Parameter	Value
Rise Delay (ps)	37.8
Fall Delay (ps)	36.9
Dynamic Energy	5.18E-09

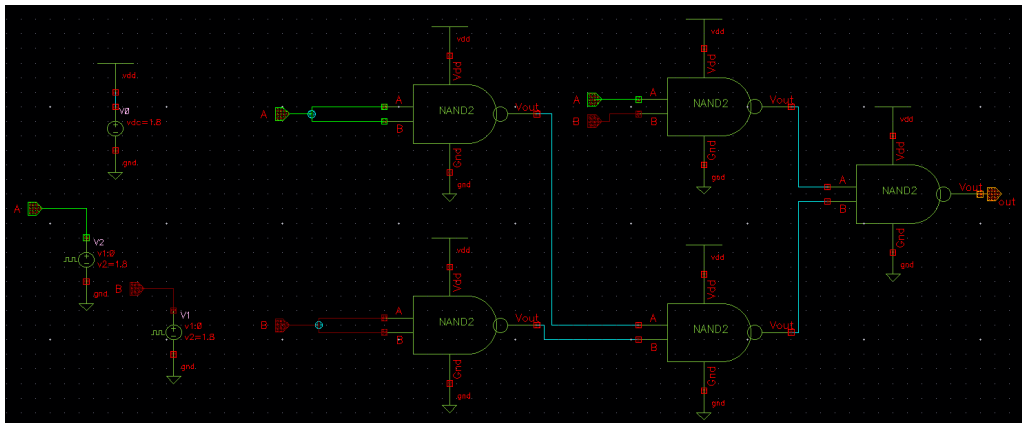


Figure 70: Schematic of XNOR Gate

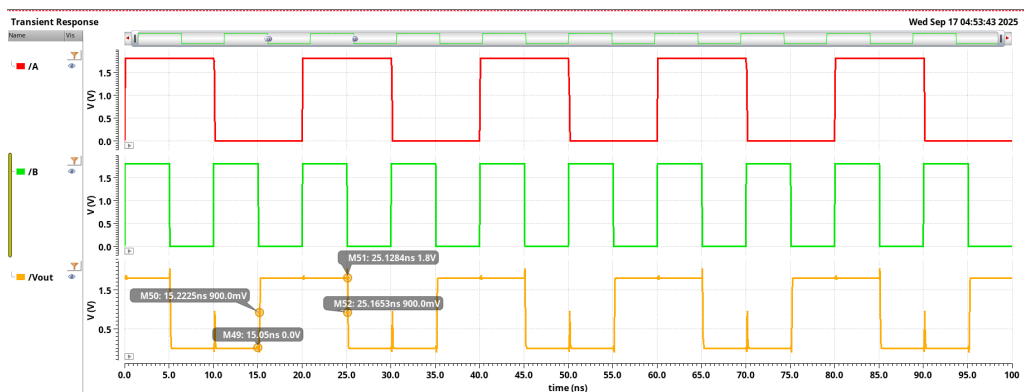


Figure 71: Rise and fall Delay

(d) $f = \overline{ab + cd}$

Table 10: Delay and Dynamic Energy Results

Parameter	Value
Rise Delay (ps)	17.4
Fall Delay (ps)	30.1
Dynamic Energy	5.66E-09

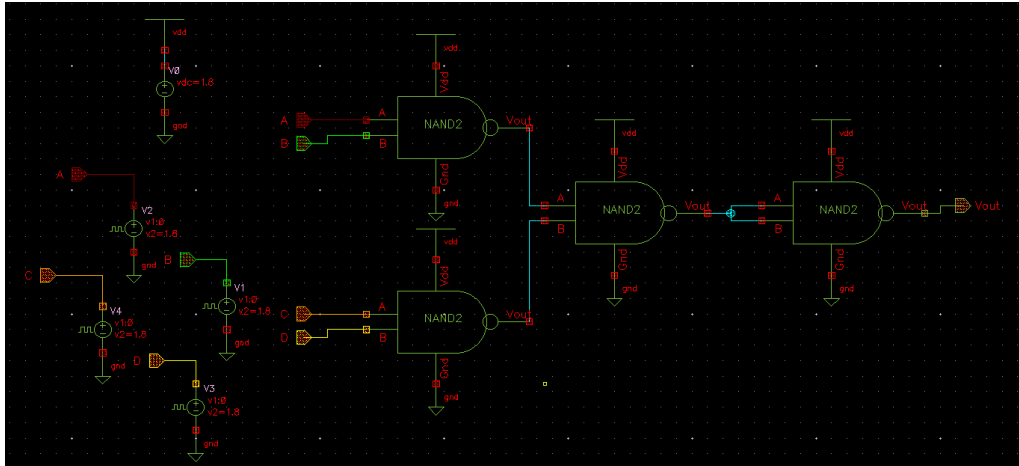
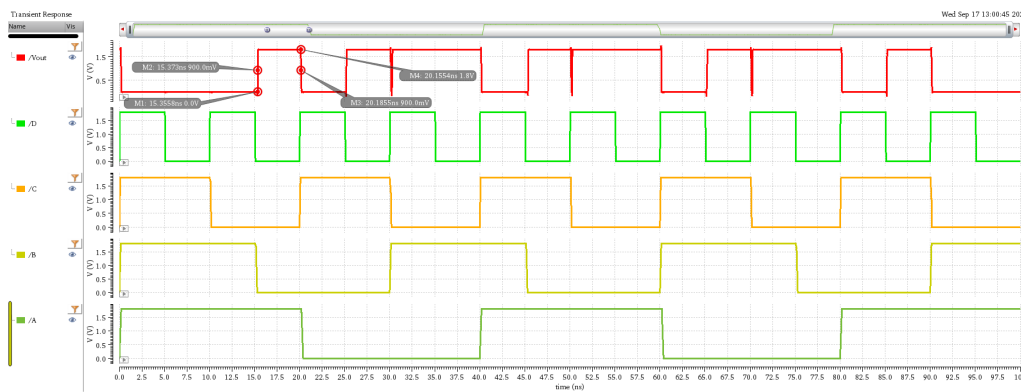
Figure 72: Schematic of $f = \overline{ab + cd}$ 

Figure 73: Rise and fall Delay

(e) $f = \overline{(a + b)cd}$

Table 11: Delay and Dynamic Energy Results

Parameter	Value
Rise Delay (ps)	37.5
Fall Delay (ps)	51
Dynamic Energy	7.34E-08

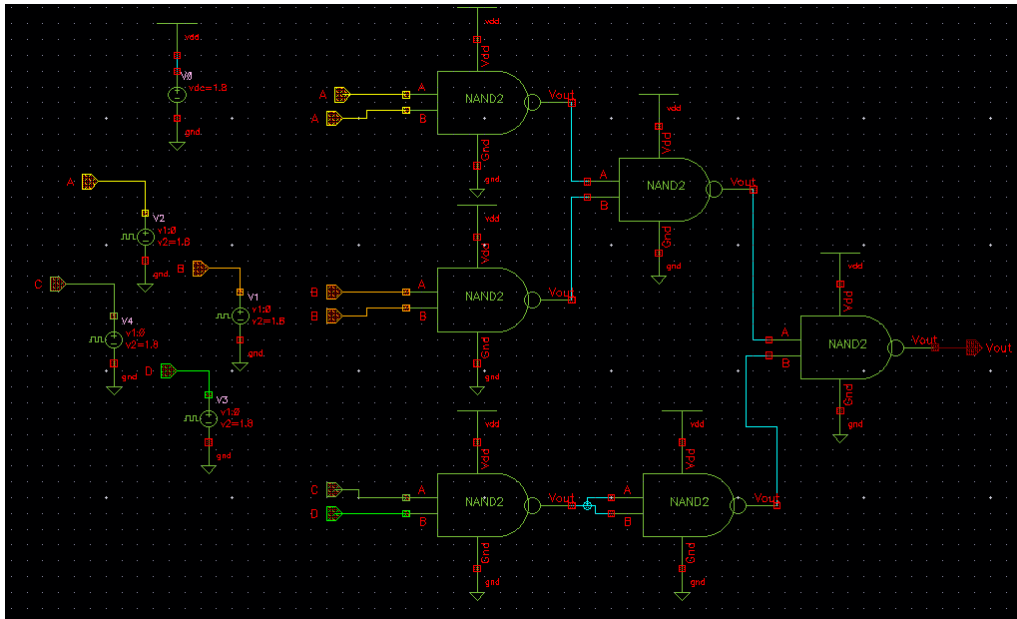
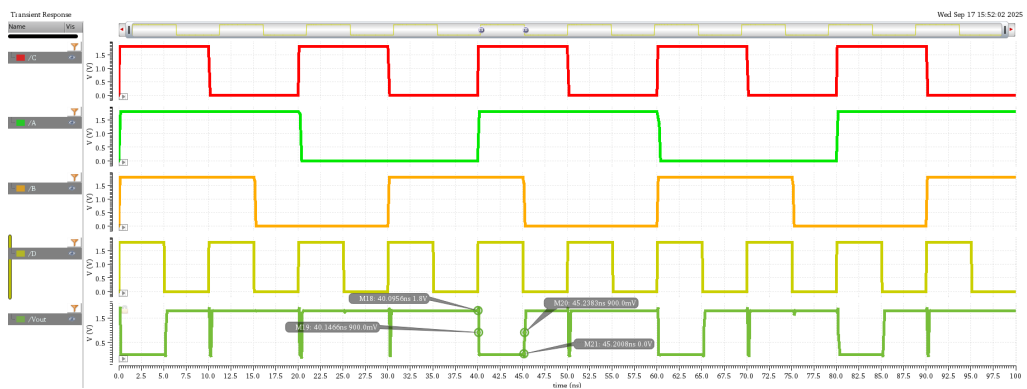
Figure 74: Schematic of $f = \overline{(a + b)cd}$ 

Figure 75: Rise and fall Delay

Question 5

Design 4-bit Ripple Carry Adder with the following designs.

- CMOS Gates (Use minimum number of logic gates)
- Mirror Adder

Compare both the designs for total dynamic energy, and delay (sum and carry).

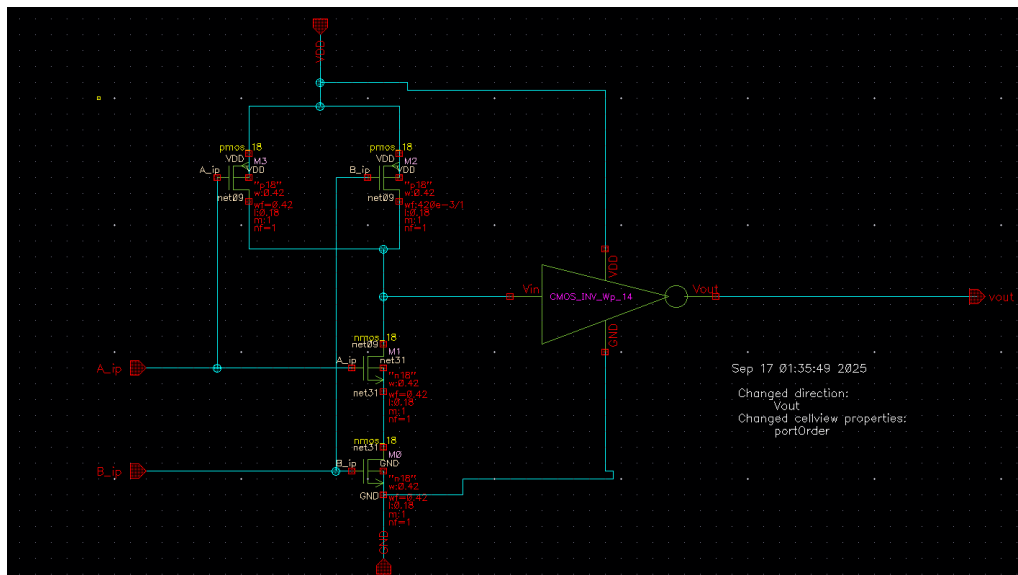


Figure 76: AND schematic

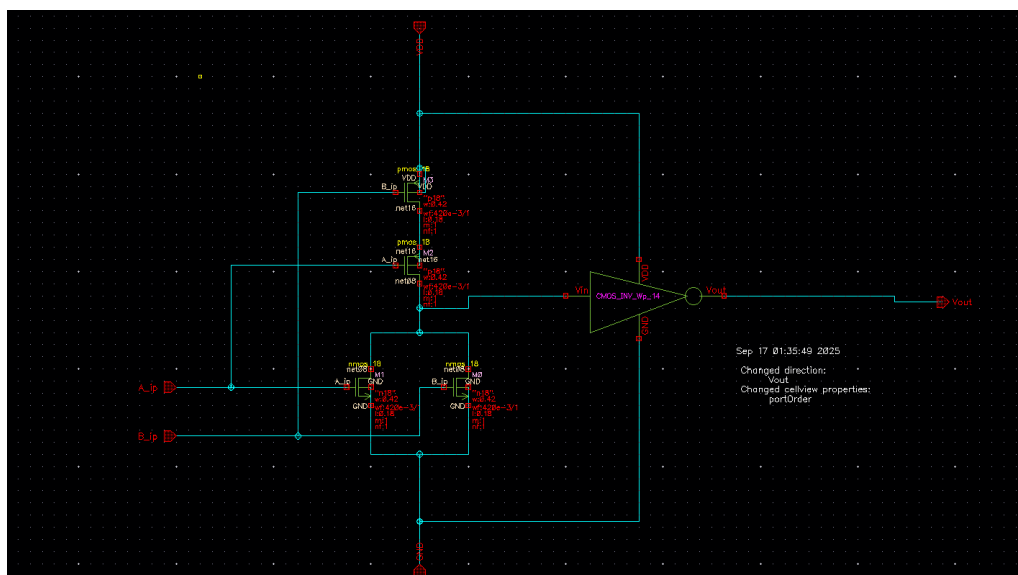


Figure 77: OR schematic

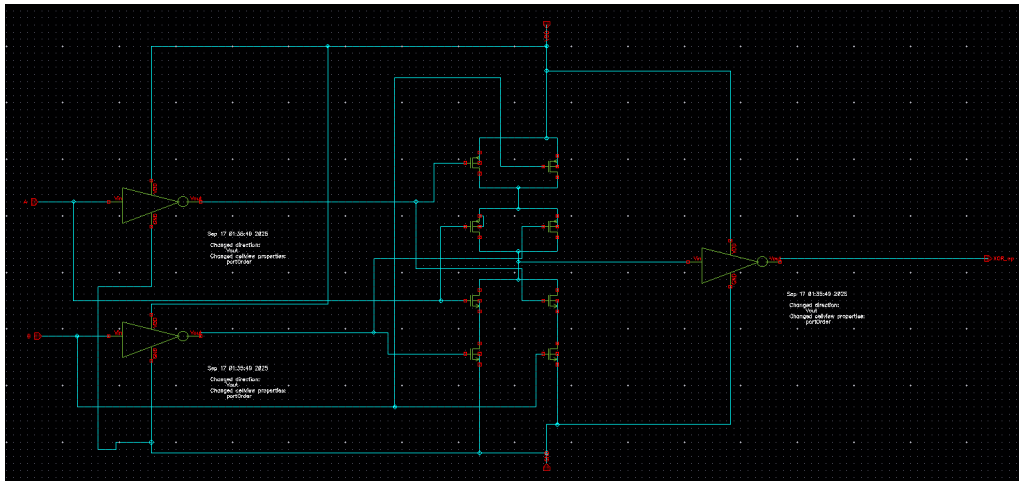


Figure 78: XOR schematic

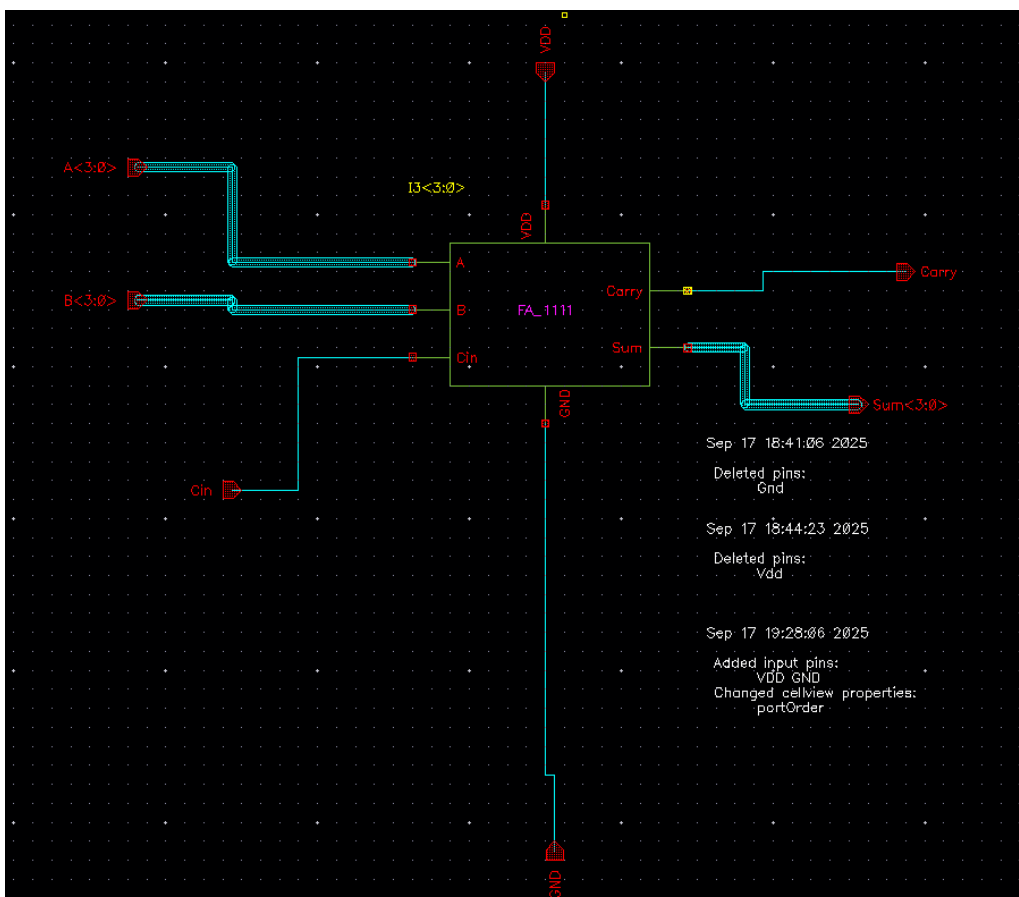


Figure 79: Ripple Carry Adder schematic

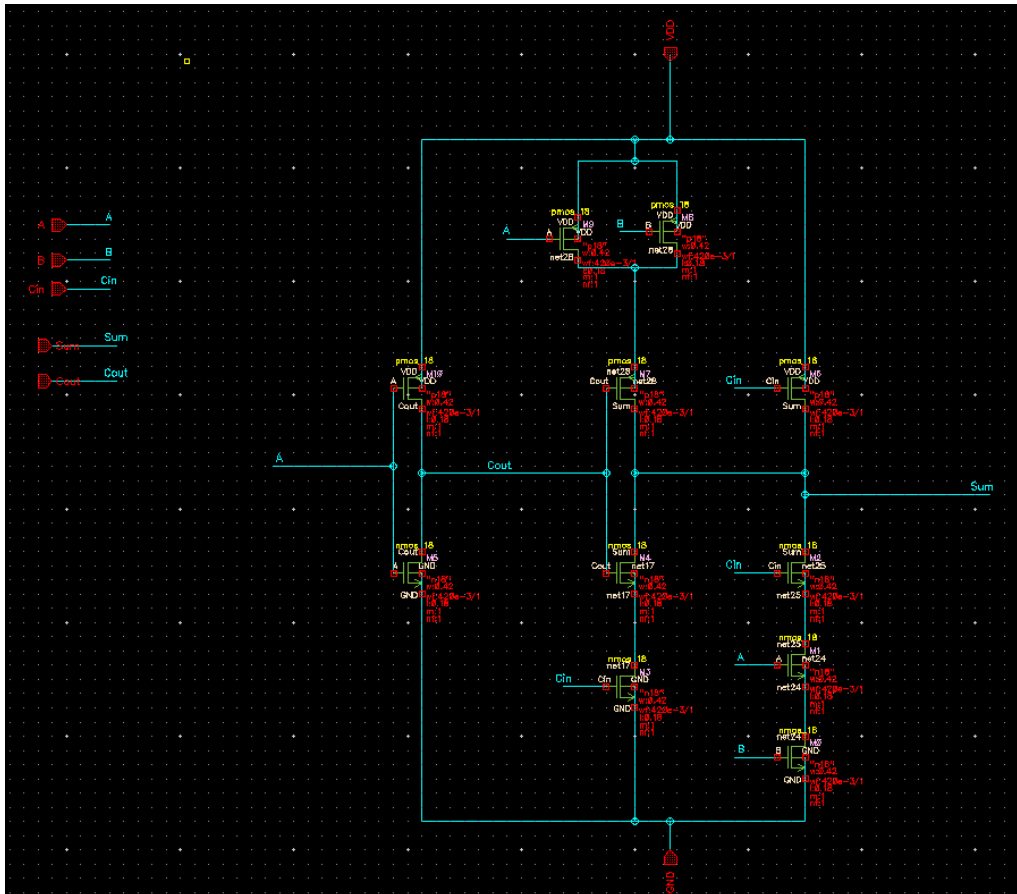


Figure 80: Mirror Adder schematic

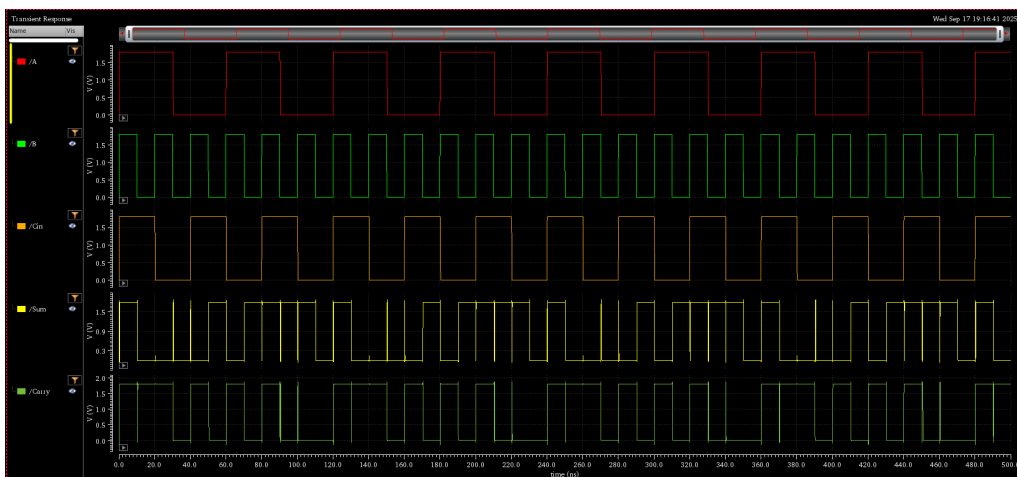


Figure 81: 4-Bit RCA Simulation

Conclusion

From this experiment, we learn about the technique and importance of making the fall time and rise time equal.